Undertow DeepSea and False Beacon Frozen Lake: physical pathology strength

Verified **336 runs and 688,128 environment episodes**. The source, configuration, frozen behavior policies, trajectories and artifact hashes are recorded per cell.

This is a paired, stationary-context finite-model experiment. It tests the deployment correction and exploration behavior; it does not establish an asymptotic coefficient-one theorem or recovery from unannounced temporal changes.

Main finding

The corrected CEAC separates its pricing-policy actor from reward-only deployment. Once the likelihood evidence supports a policy that is reward-optimal for every plausible model, free episodes use that certified policy. Statistical and Graves–Lai debts retain priority.

This study measures cumulative regret and diagnostic/nuisance visits under its recorded settings. Preflight verifies confounding and useful diagnostic routes. Sensitivity is assessed only for matched multilevel studies; no algorithm ranking is assumed in advance.

48 CEAC cells finish with a valid deployment certificate. This certifies the deployed response on the model-confidence event; it does not mean the stochastic proposal actor has converged.

The offline oracle knows the realized mode, solves its full-policy Lai–Graves allocation once, and keeps that allocation fixed throughout execution. Its privilege is explicit; it is a benchmark, not an online learner.

Where CEAC performs better

Positive saved regret means comparator minus CEAC. The additional advantage subtracts the same contrast at the mildest dose, within each seed. It separates an advantage already present at mild pathology from one that grows with pathology. All uncertainties below are sample SD across paired seeds, not confidence intervals. Bold entries favor CEAC in every recorded pair; this is descriptive, not a significance test.

False Beacon currently varies the probability of one dwell step, not beacon breadth. The full-support and excluded-support Bayesian treatments remain distinct. CEAC/IDS have candidate-model access; the primitive controls learn from transitions. These are finite-budget comparisons, not evidence that a learned evidence critic caused the advantage.

Undertow DeepSea · mode 0 · context 0

CEAC cumulative regret: 0.478 at rho=0.05 to 1.159 at rho=0.75; strongest-dose last-window value gap: 0.0000.

Full-support IDS has lower cumulative regret (0.122) at the strongest dose; its last-window value gap is 0.0000.

Strongest-dose comparison (rho=0.75); positive numbers favor CEAC.

Comparator	Regret saved	Additional advantage over mild dose	Last value-gap reduction	Pairs: CEAC lower / tied / higher regret
UCB Q-learning (Jin et al.)	**+1032.27 ± 5.69**	-6.71 ± 68.39	**+0.5044 ± 0.0048**	3 / 0 / 0
Optimistic POMD (softmax)	**+1032.88 ± 0.00**	**+26.72 ± 0.00**	**+0.5049 ± 0.0000**	3 / 0 / 0
IDAC (frequentist actor-critic)	**+1036.34 ± 2.66**	**+16.33 ± 9.60**	**+0.5066 ± 0.0013**	3 / 0 / 0
Bayesian IDS: full support	-1.04 ± 0.01	-0.68 ± 0.00	+0.0000 ± 0.0000	0 / 0 / 3

Comparator	Regret saved	Additional advantage over mild dose	Last value-gap reduction	Pairs: CEAC lower / tied / higher regret
Bayesian IDS: misspecified prior	-1.16 ± 0.00	-0.68 ± 0.00	+0.0000 ± 0.0000	0 / 0 / 3
Offline Graves–Lai oracle	-0.96 ± 0.00	-0.68 ± 0.00	+0.0000 ± 0.0000	0 / 0 / 3

Undertow DeepSea · mode 1 · context 0

CEAC cumulative regret: 0.641 at rho=0.05 to 1.365 at rho=0.75; strongest-dose last-window value gap: 0.0000.

Full-support IDS has lower cumulative regret (0.088) at the strongest dose; its last-window value gap is 0.0000.

Strongest-dose comparison (rho=0.75); positive numbers favor CEAC.

Comparator	Regret saved	Additional advantage over mild dose	Last value-gap reduction	Pairs: CEAC lower / tied / higher regret
UCB Q-learning (Jin et al.)	**+1129.61 ± 5.05**	+48.02 ± 44.15	**+0.5520 ± 0.0031**	3 / 0 / 0
Optimistic POMD (softmax)	**+1126.19 ± 0.30**	**+39.42 ± 0.07**	**+0.5506 ± 0.0000**	3 / 0 / 0
IDAC (frequentist actor-critic)	**+1131.62 ± 1.68**	**+35.21 ± 13.56**	**+0.5533 ± 0.0006**	3 / 0 / 0
Bayesian IDS: full support	-1.28 ± 0.32	-0.72 ± 0.07	+0.0000 ± 0.0000	0 / 0 / 3
Bayesian IDS: misspecified prior	**+101.03 ± 0.30**	-0.72 ± 0.07	**+0.0500 ± 0.0000**	3 / 0 / 0
Offline Graves–Lai oracle	-1.37 ± 0.30	-0.72 ± 0.07	+0.0000 ± 0.0000	0 / 0 / 3

False Beacon Frozen Lake · mode 0 · context 0

CEAC cumulative regret: 1.114 at rho=0.05 to 1.114 at rho=0.75; strongest-dose last-window value gap: 0.0000.

Full-support IDS has lower cumulative regret (0.147) at the strongest dose; its last-window value gap is 0.0000.

Strongest-dose comparison (rho=0.75); positive numbers favor CEAC.

Comparator	Regret saved	Additional advantage over mild dose	Last value-gap reduction	Pairs: CEAC lower / tied / higher regret
UCB Q-learning (Jin et al.)	**+1069.70 ± 5.71**	+2.12 ± 3.67	**+0.5235 ± 0.0031**	3 / 0 / 0
Optimistic POMD (softmax)	**+1051.74 ± 0.00**	**+0.06 ± 0.00**	**+0.5141 ± 0.0000**	3 / 0 / 0

Comparator	Regret saved	Additional advantage over mild dose	Last value-gap reduction	Pairs: CEAC lower / tied / higher regret
IDAC (frequentist actor-critic)	**+1051.90 ± 0.88**	-0.00 ± 0.11	**+0.5142 ± 0.0005**	3 / 0 / 0
Bayesian IDS: full support	-0.97 ± 0.02	-0.00 ± 0.00	+0.0000 ± 0.0000	0 / 0 / 3
Bayesian IDS: misspecified prior	-1.11 ± 0.00	-0.00 ± 0.00	+0.0000 ± 0.0000	0 / 0 / 3
Offline Graves–Lai oracle	-0.89 ± 0.00	-0.00 ± 0.00	+0.0000 ± 0.0000	0 / 0 / 3

False Beacon Frozen Lake · mode 1 · context 0

CEAC cumulative regret: 1.554 at rho=0.05 to 1.555 at rho=0.75; strongest-dose last-window value gap: 0.0000.

Full-support IDS has lower cumulative regret (0.064) at the strongest dose; its last-window value gap is 0.0000.

Strongest-dose comparison (rho=0.75); positive numbers favor CEAC.

Comparator	Regret saved	Additional advantage over mild dose	Last value-gap reduction	Pairs: CEAC lower / tied / higher regret
UCB Q-learning (Jin et al.)	**+1171.66 ± 5.22**	+2.12 ± 3.67	**+0.5735 ± 0.0031**	3 / 0 / 0
Optimistic POMD (softmax)	**+1153.67 ± 0.63**	**+0.06 ± 0.00**	**+0.5641 ± 0.0000**	3 / 0 / 0
IDAC (frequentist actor-critic)	**+1154.15 ± 1.44**	-0.05 ± 0.17	**+0.5643 ± 0.0008**	3 / 0 / 0
Bayesian IDS: full support	-1.49 ± 0.63	-0.00 ± 0.00	+0.0000 ± 0.0000	0 / 0 / 3
Bayesian IDS: misspecified prior	**+100.84 ± 0.63**	-0.00 ± 0.00	**+0.0500 ± 0.0000**	3 / 0 / 0
Offline Graves–Lai oracle	-1.56 ± 0.63	-0.00 ± 0.00	+0.0000 ± 0.0000	0 / 0 / 3

CEAC, certified deployment, and the offline oracle

The plotted CEAC is the maintained **single episode-loop, exact-model specialization**. It retains actor/reward-critic/evidence-critic updates, policy-labeled trajectory history, target/calibration, design refreshes and cumulative execution debt. Exact finite-model evaluation and separation provide its design and deployment responses. The label does not mean that the learned actor alone produced the plotted behavior.

Property	CEAC (exact-model single loop)	Offline Graves–Lai oracle
Realized mode	Unknown; inferred from real trajectories	Supplied before the first episode

Property	CEAC (exact-model single loop)	Offline Graves–Lai oracle
Candidate-model laws	Known, with exact evaluation/planning	The same laws, plus the true-mode label
Exploration allocation	Refreshed from confidence, calibration and evidence debt	Solved once for the true mode; normalized allocation stays fixed
Exploitation	Reward-only response passing a confidence-set certificate	True-mode optimal policy from the outset
Physical episodes	Statistical requests, then design debt, then certified deployment or actor	Fixed-design debt, otherwise true-mode optimal deployment

Certified deployment means the selected reward-only policy satisfies

$$\max_{m \in \mathcal{C}_t} \{J_m^\star - J_m(\widehat{\pi}_t)\} \le \tau,$$

where $\mathcal{C}_t$ is CEAC's trajectory-likelihood confidence set and $\tau = 10^{-10}$ in these runs. Conditional on the true model belonging to that set, the same bound holds for its true-model value gap. The set has a per-run anytime guarantee under the declared model family; the certificate is not a guarantee under arbitrary model misspecification.

Certification is checked without the true-mode label. It affects free-episode exploitation; statistical and design debts still run first. The oracle also executes scheduled experiments even though it knows the true mode, so its plotted finite-budget cost is the implemented fixed-design benchmark, not a finite-time lower bound every method must exceed.

The exact tabular separator and Appendix D's reward-only response are part of the v90 finite-model framework. Connecting that certified response to free-episode execution is the documented implementation refinement. This preserves one environment episode loop while using exact optimization at refreshes; it does not establish actor/critic convergence or a literal empirical-only implementation of every pseudocode estimator.

Comparator access and scope

Method	Information and policy access
CEAC (exact-model single loop)	Unknown realized mode; exact planning over declared models; certified reward-only execution
Offline Graves–Lai oracle	Realized model known; allocation computed offline and fixed
Bayesian IDS: full support	Declared model family; full-support Bayesian IDS; full policy response
IDAC (frequentist actor-critic)	Primitive trajectories only; upstream tabular softmax actor and two critics
UCB Q-learning (Jin et al.)	Primitive trajectories only; Jin et al. Algorithm 1 UCB-Hoeffding Q-learning
Optimistic POMD (softmax)	Primitive trajectories only; optimistic POMD softmax policy and learned empirical model
Bayesian IDS: misspecified prior	Same IDS, with shifted mode explicitly excluded from prior support

Physical pathology strength: matched method overlays

The standard figures overlay UCB-Hoeffding Q-learning, optimistic softmax POMD, IDAC (frequentist occupancy-information actor-critic), Bayesian IDS with full support, Bayesian IDS with a misspecified prior, CEAC and the offline oracle at each strength. Both Bayesian arms use the same algorithm: full support is p0=(0.5,0.5);

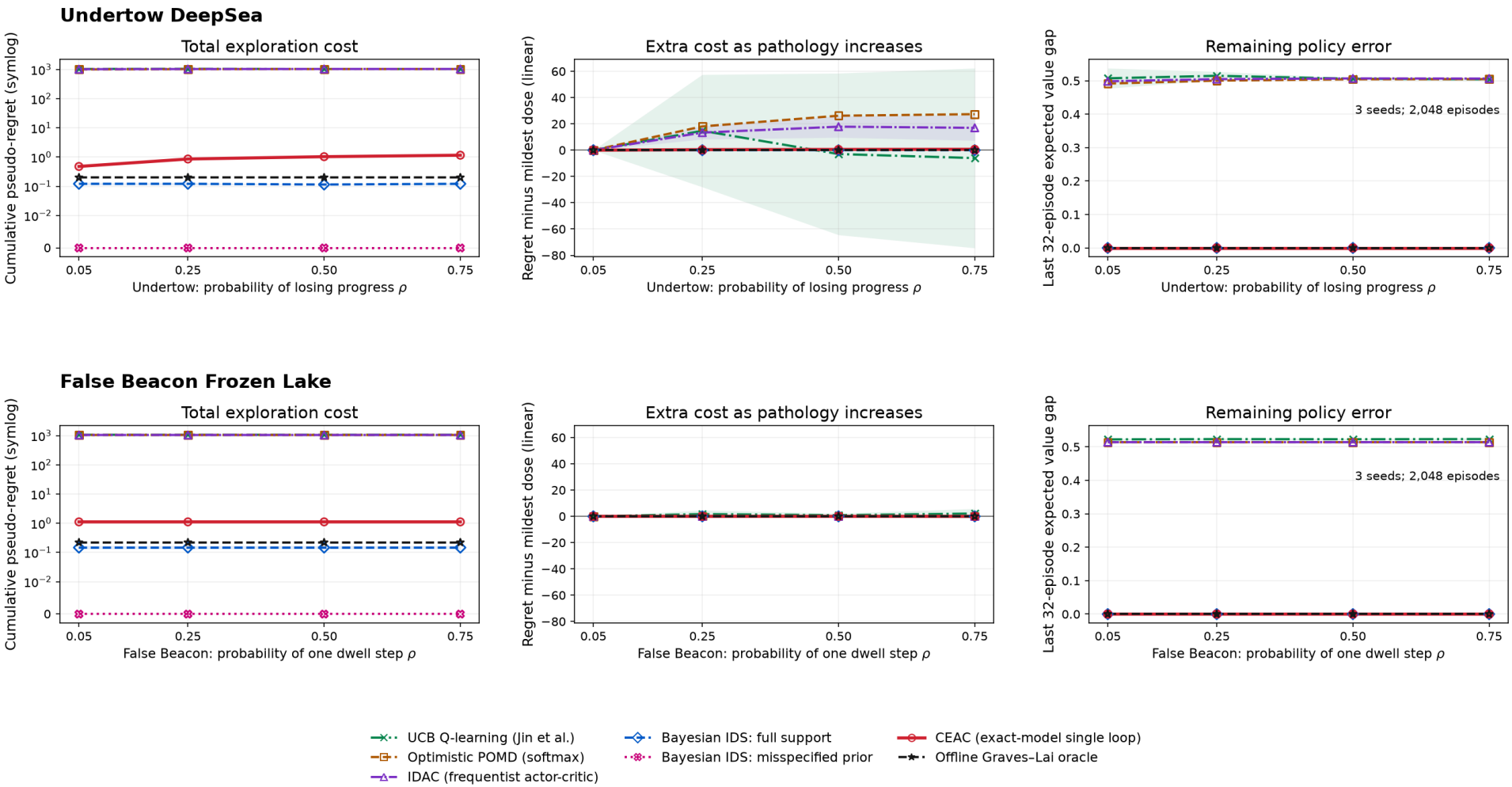


Figure 1: pathology response context0 mode0

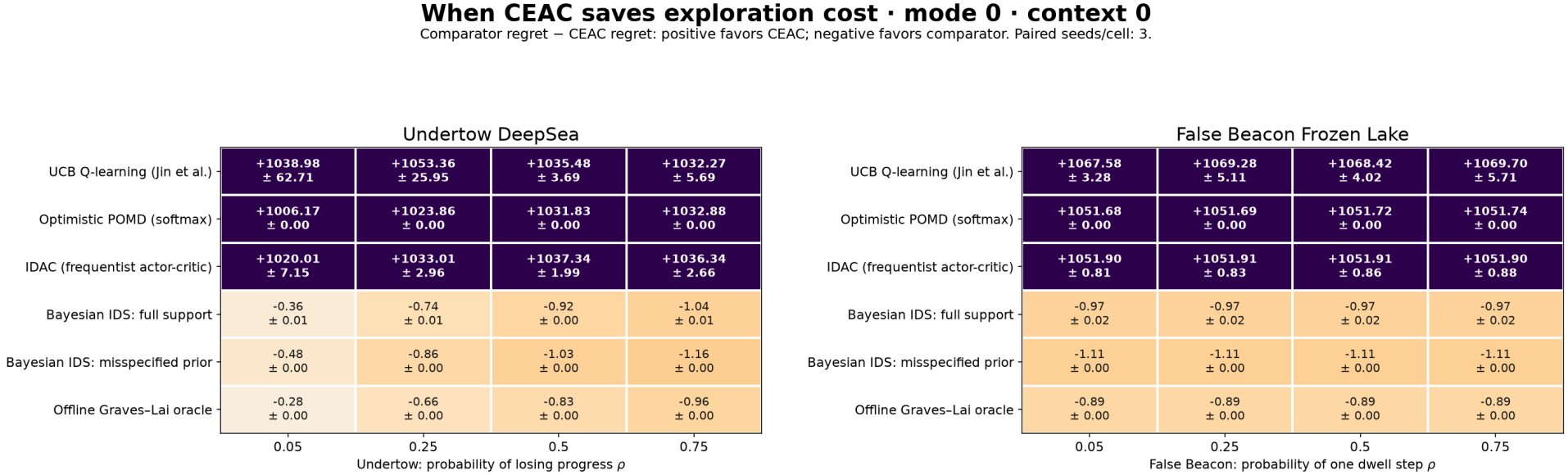


Figure 2: pathology ceac advantage context0 mode0

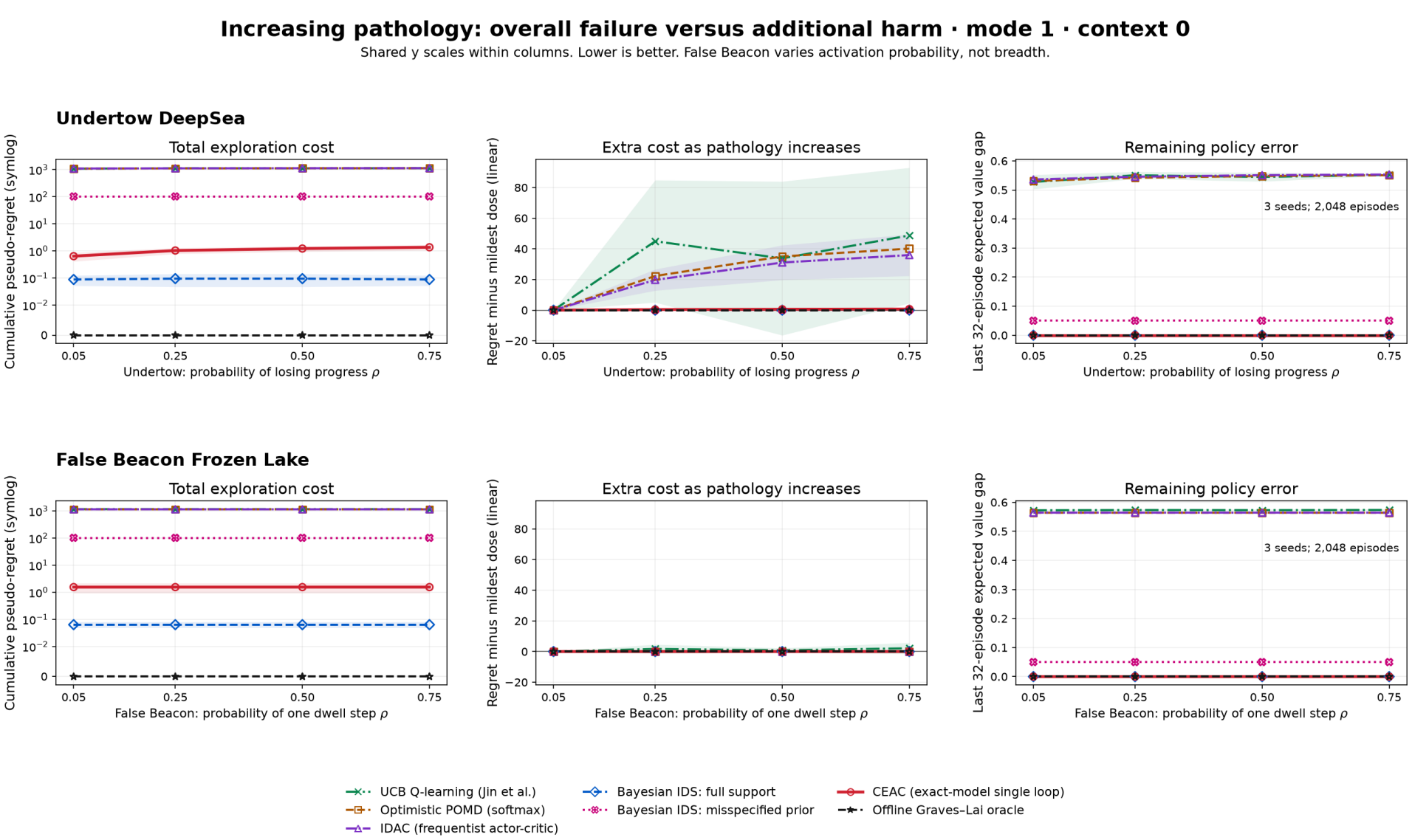


Figure 3: pathology response context0 mode1

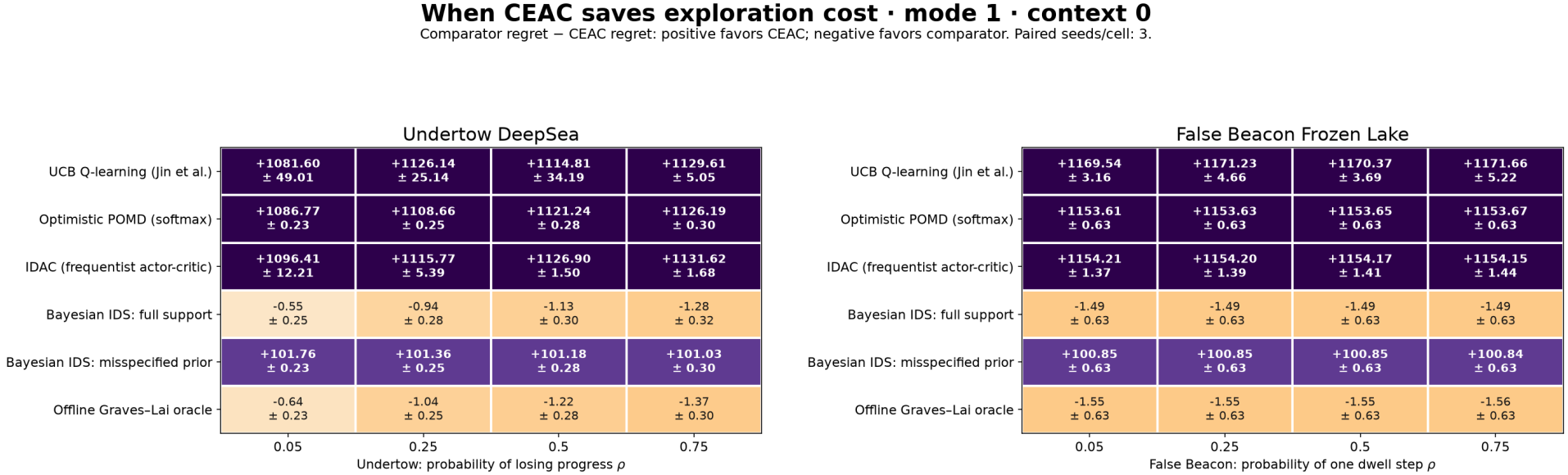


Figure 4: pathology ceac advantage context0 mode1

the restricted prior is p0=(1,0). The latter excludes the true model in mode 1; mode 0 is its correctly specified negative control. Prior support is held fixed while physical diversion strength varies. IDAC optimizes an occupancy-entropy ratio; it is not a frequentist implementation of the same posterior IDS objective.

Only the mode-independent diversion probability rho changes (0.05, 0.25, 0.50, 0.75). Interior probabilities preserve the same reachable state/action domain. Geometry, horizon, four nuisance symbols, useful-route values, probe KL and reward ranges are checked across settings before training. A DeepSea nuisance move loses rightward progress with probability rho; otherwise it advances. On the first reed visit, False Beacon adds one forced physical dwell step with probability rho. Both branches have zero task KL. The beacon state records pending/completed dwell; seed routes resume their intended leg after the wait.

$$P_{m,\rho}(\cdot \mid s, a_N) = (1-\rho)P_{\text{pass}}(\cdot \mid s, a_N) + \rho P_{\text{divert}}(\cdot \mid s, a_N).$$

The report automatically emits method overlays, STEERING-style strength curves in separate method panels, paired added-regret traces, and exposure/value/success diagnostics. Training traces use shared symlog axes, linear within ± 0.01, to keep small costs visible. Markers are drawn above the lines on staggered recorded episodes; no values are shifted. A strongest-dose endpoint view gives each method its own row, including exact ties. Raw per-seed summaries and paired added regret are saved in pathology_statistics.json. No monotone deterioration or algorithm ranking is imposed. Training budgets match in episodes; physical steps are recorded separately. CEAC/IDS have candidate-model access that the primitive controls lack.

Environment	Mode	Method	Regret: mild / strong	Paired added regret (mean ± sample SD)	Last value gap: mild / strong	POMD all-zero costs, strong
deepsea (c0)	0	CEAC (exact-model single loop)	0.478 / 1.159	0.682 ± 0.000	0.0000 / 0.0000	n/a
deepsea (c0)	0	IDAC (frequentist actor-critic)	1020.493 / 1037.500	17.007 ± 9.598	0.4985 / 0.5066	n/a
deepsea (c0)	0	Bayesian IDS: full support	0.122 / 0.122	0.000 ± 0.000	0.0000 / 0.0000	n/a
deepsea (c0)	0	Bayesian IDS: misspecified prior	0.000 / 0.000	0.000 ± 0.000	0.0000 / 0.0000	n/a
deepsea (c0)	0	Offline Graves–Lai oracle	0.200 / 0.200	0.000 ± 0.000	0.0000 / 0.0000	n/a
deepsea (c0)	0	Optimistic POMD (softmax)	1006.645 / 1034.043	27.398 ± 0.000	0.4915 / 0.5049	100.0%
deepsea (c0)	0	UCB Q-learning (Jin et al.)	1039.457 / 1033.430	-6.028 ± 68.390	0.5073 / 0.5044	n/a
deepsea (c0)	1	CEAC (exact-model single loop)	0.641 / 1.365	0.724 ± 0.074	0.0000 / 0.0000	n/a
deepsea (c0)	1	IDAC (frequentist actor-critic)	1097.053 / 1132.990	35.937 ± 13.519	0.5360 / 0.5533	n/a
deepsea (c0)	1	Bayesian IDS: full support	0.088 / 0.088	0.000 ± 0.000	0.0000 / 0.0000	n/a

Environment	Mode	Method	Regret: mild / strong	Paired added regret (mean ± sample SD)	Last value gap: mild / strong	POMD all-zero costs, strong
deepsea (c0)	1	Bayesian IDS: misspecified prior	102.400 / 102.400	0.000 ± 0.000	0.0500 / 0.0500	n/a
deepsea (c0)	1	Offline Graves–Lai oracle	0.000 / 0.000	0.000 ± 0.000	0.0000 / 0.0000	n/a
deepsea (c0)	1	Optimistic POMD (softmax)	1087.413 / 1127.554	40.141 ± 0.000	0.5310 / 0.5506	100.0%
deepsea (c0)	1	UCB Q-learning (Jin et al.)	1082.237 / 1130.977	48.740 ± 44.195	0.5276 / 0.5520	n/a
frozenlake (c0)	0	CEAC (exact-model single loop)	1.114 / 1.114	0.001 ± 0.000	0.0000 / 0.0000	n/a
frozenlake (c0)	0	IDAC (frequentist actor-critic)	1053.018 / 1053.016	-0.002 ± 0.111	0.5142 / 0.5142	n/a
frozenlake (c0)	0	Bayesian IDS: full support	0.147 / 0.147	0.000 ± 0.000	0.0000 / 0.0000	n/a
frozenlake (c0)	0	Bayesian IDS: misspecified prior	0.000 / 0.000	0.000 ± 0.000	0.0000 / 0.0000	n/a
frozenlake (c0)	0	Offline Graves–Lai oracle	0.220 / 0.220	0.000 ± 0.000	0.0000 / 0.0000	n/a
frozenlake (c0)	0	Optimistic POMD (softmax)	1052.790 / 1052.851	0.062 ± 0.000	0.5141 / 0.5141	100.0%
frozenlake (c0)	0	UCB Q-learning (Jin et al.)	1068.693 / 1070.813	2.120 ± 3.672	0.5225 / 0.5235	n/a
frozenlake (c0)	1	CEAC (exact-model single loop)	1.554 / 1.555	0.001 ± 0.000	0.0000 / 0.0000	n/a
frozenlake (c0)	1	IDAC (frequentist actor-critic)	1155.760 / 1155.707	-0.054 ± 0.171	0.5643 / 0.5643	n/a
frozenlake (c0)	1	Bayesian IDS: full support	0.064 / 0.064	0.000 ± 0.000	0.0000 / 0.0000	n/a
frozenlake (c0)	1	Bayesian IDS: misspecified prior	102.400 / 102.400	0.000 ± 0.000	0.0500 / 0.0500	n/a
frozenlake (c0)	1	Offline Graves–Lai oracle	0.000 / 0.000	0.000 ± 0.000	0.0000 / 0.0000	n/a

Environment	Mode	Method	Regret: mild / strong	Paired added regret (mean ± sample SD)	Last value gap: mild / strong	POMD all-zero costs, strong
frozenlake (c0)	1	Optimistic POMD (softmax)	1155.167 / 1155.229	0.062 ± 0.000	0.5640 / 0.5641	100.0%
frozenlake (c0)	1	UCB Q-learning (Jin et al.)	1171.091 / 1173.212	2.120 ± 3.673	0.5725 / 0.5735	n/a

Diagnostic flags describe observations, not causal proofs: POMD_BONUS_SATURATION means at least 90% of episodes had all optimistic costs zero; PHYSICAL_NUISANCE_EXPOSURE records lost progress/dwell; FINITE_BUDGET_VALUE_GAP means a final 32-episode mean expected gap above 0.01. A plateau already present at mild strength is not evidence that increasing pathology caused it. PRIOR_SUPPORT_EXCLUDED is the deliberately misspecified zero-prior treatment, explicitly labeled alongside full-support Bayesian IDS in every main figure. A dedicated prior comparison also isolates that assumption.

Prior support: delayed discovery versus permanent exclusion

The recorded IDS priors are listed below. The zero arm is an explicit misspecified Bayesian support treatment. CEAC and full-support IDS retain both candidate modes; this is an information-assumption difference, not a claim that CEAC can identify a model outside its own candidate family.

For positive observation likelihoods, Bayesian updating preserves zero prior mass:

$$p_t(1) = \frac{p_0(1)L_t(1)}{p_0(0)L_t(0) + p_0(1)L_t(1)} = 0 \quad \text{if } p_0(1) = 0.$$

The selected old-route trajectory law is identical across modes. With mode 1 realized, its positive value gap is Delta=e. IDS assigns this old route zero subjective regret and zero information; under the declared IDS zero/zero convention it never requests a diagnostic. The exact prediction is R_T=T Delta, using each recorded episode budget and kernel value gap. The audit checks the prediction at every episode, and also checks the mode-0 negative control. A small positive prior is different: it can recover, but diagnostic waiting times may exceed this finite budget.

Setting	Environment	Mode	IDS prior	Seeds	Final posterior for mode 1	Cumulative regret	Diagnostic visits
pathology_rho005	Undertow DeepSea	0	0	3	0	0.0000	0.0
pathology_rho005	Undertow DeepSea	0	0.5	3	0.000124143	0.1217	5.3
pathology_rho005	Undertow DeepSea	1	0	3	0	102.4000	0.0
pathology_rho005	Undertow DeepSea	1	0.5	3	1	0.0883	3.7
pathology_rho005	False Beacon Frozen Lake	0	0	3	0	0.0000	0.0
pathology_rho005	False Beacon Frozen Lake	0	0.5	3	0.000124143	0.1470	6.0

Setting	Environment	Mode	IDS prior	Seeds	Final posterior for mode 1	Cumulative regret	Diagnostic visits
pathology_rho005	False Beacon Frozen Lake	1	0	3	0	102.4000	0.0
pathology_rho005	False Beacon Frozen Lake	1	0.5	3	1	0.0638	2.3
pathology_rho025	Undertow DeepSea	0	0	3	0	0.0000	0.0
pathology_rho025	Undertow DeepSea	0	0.5	3	0.000124143	0.1217	5.3
pathology_rho025	Undertow DeepSea	1	0	3	0	102.4000	0.0
pathology_rho025	Undertow DeepSea	1	0.5	3	1	0.0950	4.0
pathology_rho025	False Beacon Frozen Lake	0	0	3	0	0.0000	0.0
pathology_rho025	False Beacon Frozen Lake	0	0.5	3	0.000124143	0.1470	6.0
pathology_rho025	False Beacon Frozen Lake	1	0	3	0	102.4000	0.0
pathology_rho025	False Beacon Frozen Lake	1	0.5	3	1	0.0638	2.3
pathology_rho050	Undertow DeepSea	0	0	3	0	0.0000	0.0
pathology_rho050	Undertow DeepSea	0	0.5	3	0.000171115	0.1150	5.0
pathology_rho050	Undertow DeepSea	1	0	3	0	102.4000	0.0
pathology_rho050	Undertow DeepSea	1	0.5	3	1	0.0950	4.0
pathology_rho050	False Beacon Frozen Lake	0	0	3	0	0.0000	0.0
pathology_rho050	False Beacon Frozen Lake	0	0.5	3	0.000124143	0.1470	6.0
pathology_rho050	False Beacon Frozen Lake	1	0	3	0	102.4000	0.0
pathology_rho050	False Beacon Frozen Lake	1	0.5	3	1	0.0638	2.3
pathology_rho075	Undertow DeepSea	0	0	3	0	0.0000	0.0
pathology_rho075	Undertow DeepSea	0	0.5	3	0.000124143	0.1217	5.3
pathology_rho075	Undertow DeepSea	1	0	3	0	102.4000	0.0

Setting	Environment	Mode	IDS prior	Seeds	Final posterior for mode 1	Cumulative regret	Diagnostic visits
pathology_rho075	Undertow DeepSea	1	0.5	3	1	0.0883	3.7
pathology_rho075	False Beacon Frozen Lake	0	0	3	0	0.0000	0.0
pathology_rho075	False Beacon Frozen Lake	0	0.5	3	0.000124143	0.1470	6.0
pathology_rho075	False Beacon Frozen Lake	1	0	3	0	102.4000	0.0
pathology_rho075	False Beacon Frozen Lake	1	0.5	3	1	0.0638	2.3

This is a static out-of-support task mode. It does not yet test an unknown change point, change-aware Bayesian filtering, posterior resets or model-class expansion.

What these traces establish about evidence and replay

Saved trajectories reconstruct the cumulative likelihood mixture and confidence sets in all 48 CEAC cells, with maximum probability residual 1.11e-15. Reported policy-credit counts match physical trajectory counts at every episode. The audit is saved in `history_audit.json`.

36 of 48 CEAC runs made zero actor/evidence-critic updates; 0 learned actor proposals were admitted. Actor and evidence-value updates occur only on episodes labeled actor. Exact declared-model evaluation supplies diagnostic-design coefficients and response/deployment certificates. These experiments therefore support cumulative evidence accounting, directed experimental design and certified deployment; they do not isolate a causal performance gain from the learned TD evidence critic or replay resampling.

The replay object retains policy-labeled trajectories and cumulative likelihood sufficient statistics. Its ledger report exposes policies with quotas/statistical requests; it does not expose all internal buffer entries. The reconstruction verifies every saved episode and every reported credit, rather than interpreting missing ledger keys as discarded evidence. No frozen diagnostic/deployment trajectory is counted as an off-policy actor update.

A learned-critic claim needs an active learned-estimator lane and matched ablations of evidence pricing and cumulative-history use. Those experiments are not inferred from these cached curves.

Primitive learner update diagnostics

These runs validate the new primitive-action implementations. A short run is not a convergence claim or evidence of systematic failure under increasing pathology strength. In particular, the published POMD bonuses can leave every optimistic cost clipped to zero, so its softmax policy stays uniform.

Setting	Environment	Mode	Method	Seed	Episodes	Actor / Q updates	Episodes with all POMD costs zero
pathology_rho005	deepsea	0	IDAC (frequentist actor-critic)	0	2048	128	n/a
pathology_rho005	deepsea	0	UCB Q-learning (Jin et al.)	0	2048	18432	n/a

Setting	Environment	Mode	Method	Seed	Episodes	Actor / Q updates	Episodes with all POMD costs zero
pathology_rho005	deepsea	0	Optimistic POMD (softmax)	0	2048	2048	2048
pathology_rho005	deepsea	0	IDAC (frequentist actor-critic)	1	2048	128	n/a
pathology_rho005	deepsea	0	UCB Q-learning (Jin et al.)	1	2048	18432	n/a
pathology_rho005	deepsea	0	Optimistic POMD (softmax)	1	2048	2048	2048
pathology_rho005	deepsea	0	IDAC (frequentist actor-critic)	2	2048	128	n/a
pathology_rho005	deepsea	0	UCB Q-learning (Jin et al.)	2	2048	18432	n/a
pathology_rho005	deepsea	0	Optimistic POMD (softmax)	2	2048	2048	2048
pathology_rho005	deepsea	1	IDAC (frequentist actor-critic)	0	2048	128	n/a
pathology_rho005	deepsea	1	UCB Q-learning (Jin et al.)	0	2048	18432	n/a
pathology_rho005	deepsea	1	Optimistic POMD (softmax)	0	2048	2048	2048
pathology_rho005	deepsea	1	IDAC (frequentist actor-critic)	1	2048	128	n/a
pathology_rho005	deepsea	1	UCB Q-learning (Jin et al.)	1	2048	18432	n/a
pathology_rho005	deepsea	1	Optimistic POMD (softmax)	1	2048	2048	2048
pathology_rho005	deepsea	1	IDAC (frequentist actor-critic)	2	2048	128	n/a

Setting	Environment	Mode	Method	Seed	Episodes	Actor / Q updates	Episodes with all POMD costs zero
pathology_rho005	deepsea	1	UCB Q-learning (Jin et al.)	2	2048	18432	n/a
pathology_rho005	deepsea	1	Optimistic POMD (softmax)	2	2048	2048	2048
pathology_rho005	frozenlake	0	IDAC (frequentist actor-critic)	0	2048	128	n/a
pathology_rho005	frozenlake	0	UCB Q-learning (Jin et al.)	0	2048	32768	n/a
pathology_rho005	frozenlake	0	Optimistic POMD (softmax)	0	2048	2048	2048
pathology_rho005	frozenlake	0	IDAC (frequentist actor-critic)	1	2048	128	n/a
pathology_rho005	frozenlake	0	UCB Q-learning (Jin et al.)	1	2048	32768	n/a
pathology_rho005	frozenlake	0	Optimistic POMD (softmax)	1	2048	2048	2048
pathology_rho005	frozenlake	0	IDAC (frequentist actor-critic)	2	2048	128	n/a
pathology_rho005	frozenlake	0	UCB Q-learning (Jin et al.)	2	2048	32768	n/a
pathology_rho005	frozenlake	0	Optimistic POMD (softmax)	2	2048	2048	2048
pathology_rho005	frozenlake	1	IDAC (frequentist actor-critic)	0	2048	128	n/a
pathology_rho005	frozenlake	1	UCB Q-learning (Jin et al.)	0	2048	32768	n/a
pathology_rho005	frozenlake	1	Optimistic POMD (softmax)	0	2048	2048	2048

Setting	Environment	Mode	Method	Seed	Episodes	Actor / Q updates	Episodes with all POMD costs zero
pathology_rho005	frozenlake	1	IDAC (frequentist actor-critic)	1	2048	128	n/a
pathology_rho005	frozenlake	1	UCB Q-learning (Jin et al.)	1	2048	32768	n/a
pathology_rho005	frozenlake	1	Optimistic POMD (softmax)	1	2048	2048	2048
pathology_rho005	frozenlake	1	IDAC (frequentist actor-critic)	2	2048	128	n/a
pathology_rho005	frozenlake	1	UCB Q-learning (Jin et al.)	2	2048	32768	n/a
pathology_rho005	frozenlake	1	Optimistic POMD (softmax)	2	2048	2048	2048
pathology_rho025	deepsea	0	IDAC (frequentist actor-critic)	0	2048	128	n/a
pathology_rho025	deepsea	0	UCB Q-learning (Jin et al.)	0	2048	18432	n/a
pathology_rho025	deepsea	0	Optimistic POMD (softmax)	0	2048	2048	2048
pathology_rho025	deepsea	0	IDAC (frequentist actor-critic)	1	2048	128	n/a
pathology_rho025	deepsea	0	UCB Q-learning (Jin et al.)	1	2048	18432	n/a
pathology_rho025	deepsea	0	Optimistic POMD (softmax)	1	2048	2048	2048
pathology_rho025	deepsea	0	IDAC (frequentist actor-critic)	2	2048	128	n/a
pathology_rho025	deepsea	0	UCB Q-learning (Jin et al.)	2	2048	18432	n/a

Setting	Environment	Mode	Method	Seed	Episodes	Actor / Q updates	Episodes with all POMD costs zero
pathology_rho025	deepsea	0	Optimistic POMD (softmax)	2	2048	2048	2048
pathology_rho025	deepsea	1	IDAC (frequentist actor-critic)	0	2048	128	n/a
pathology_rho025	deepsea	1	UCB Q-learning (Jin et al.)	0	2048	18432	n/a
pathology_rho025	deepsea	1	Optimistic POMD (softmax)	0	2048	2048	2048
pathology_rho025	deepsea	1	IDAC (frequentist actor-critic)	1	2048	128	n/a
pathology_rho025	deepsea	1	UCB Q-learning (Jin et al.)	1	2048	18432	n/a
pathology_rho025	deepsea	1	Optimistic POMD (softmax)	1	2048	2048	2048
pathology_rho025	deepsea	1	IDAC (frequentist actor-critic)	2	2048	128	n/a
pathology_rho025	deepsea	1	UCB Q-learning (Jin et al.)	2	2048	18432	n/a
pathology_rho025	deepsea	1	Optimistic POMD (softmax)	2	2048	2048	2048
pathology_rho025	frozenlake	0	IDAC (frequentist actor-critic)	0	2048	128	n/a
pathology_rho025	frozenlake	0	UCB Q-learning (Jin et al.)	0	2048	32768	n/a
pathology_rho025	frozenlake	0	Optimistic POMD (softmax)	0	2048	2048	2048
pathology_rho025	frozenlake	0	IDAC (frequentist actor-critic)	1	2048	128	n/a

Setting	Environment	Mode	Method	Seed	Episodes	Actor / Q updates	Episodes with all POMD costs zero
pathology_rho025	frozenlake	0	UCB Q-learning (Jin et al.)	1	2048	32768	n/a
pathology_rho025	frozenlake	0	Optimistic POMD (softmax)	1	2048	2048	2048
pathology_rho025	frozenlake	0	IDAC (frequentist actor-critic)	2	2048	128	n/a
pathology_rho025	frozenlake	0	UCB Q-learning (Jin et al.)	2	2048	32768	n/a
pathology_rho025	frozenlake	0	Optimistic POMD (softmax)	2	2048	2048	2048
pathology_rho025	frozenlake	1	IDAC (frequentist actor-critic)	0	2048	128	n/a
pathology_rho025	frozenlake	1	UCB Q-learning (Jin et al.)	0	2048	32768	n/a
pathology_rho025	frozenlake	1	Optimistic POMD (softmax)	0	2048	2048	2048
pathology_rho025	frozenlake	1	IDAC (frequentist actor-critic)	1	2048	128	n/a
pathology_rho025	frozenlake	1	UCB Q-learning (Jin et al.)	1	2048	32768	n/a
pathology_rho025	frozenlake	1	Optimistic POMD (softmax)	1	2048	2048	2048
pathology_rho025	frozenlake	1	IDAC (frequentist actor-critic)	2	2048	128	n/a
pathology_rho025	frozenlake	1	UCB Q-learning (Jin et al.)	2	2048	32768	n/a
pathology_rho025	frozenlake	1	Optimistic POMD (softmax)	2	2048	2048	2048

Setting	Environment	Mode	Method	Seed	Episodes	Actor / Q updates	Episodes with all POMD costs zero
pathology_rho050	deepsea	0	IDAC (frequentist actor-critic)	0	2048	128	n/a
pathology_rho050	deepsea	0	UCB Q-learning (Jin et al.)	0	2048	18432	n/a
pathology_rho050	deepsea	0	Optimistic POMD (softmax)	0	2048	2048	2048
pathology_rho050	deepsea	0	IDAC (frequentist actor-critic)	1	2048	128	n/a
pathology_rho050	deepsea	0	UCB Q-learning (Jin et al.)	1	2048	18432	n/a
pathology_rho050	deepsea	0	Optimistic POMD (softmax)	1	2048	2048	2048
pathology_rho050	deepsea	0	IDAC (frequentist actor-critic)	2	2048	128	n/a
pathology_rho050	deepsea	0	UCB Q-learning (Jin et al.)	2	2048	18432	n/a
pathology_rho050	deepsea	0	Optimistic POMD (softmax)	2	2048	2048	2048
pathology_rho050	deepsea	1	IDAC (frequentist actor-critic)	0	2048	128	n/a
pathology_rho050	deepsea	1	UCB Q-learning (Jin et al.)	0	2048	18432	n/a
pathology_rho050	deepsea	1	Optimistic POMD (softmax)	0	2048	2048	2048
pathology_rho050	deepsea	1	IDAC (frequentist actor-critic)	1	2048	128	n/a
pathology_rho050	deepsea	1	UCB Q-learning (Jin et al.)	1	2048	18432	n/a

Setting	Environment	Mode	Method	Seed	Episodes	Actor / Q updates	Episodes with all POMD costs zero
pathology_rho050	deepsea	1	Optimistic POMD (softmax)	1	2048	2048	2048
pathology_rho050	deepsea	1	IDAC (frequentist actor-critic)	2	2048	128	n/a
pathology_rho050	deepsea	1	UCB Q-learning (Jin et al.)	2	2048	18432	n/a
pathology_rho050	deepsea	1	Optimistic POMD (softmax)	2	2048	2048	2048
pathology_rho050	frozenlake	0	IDAC (frequentist actor-critic)	0	2048	128	n/a
pathology_rho050	frozenlake	0	UCB Q-learning (Jin et al.)	0	2048	32768	n/a
pathology_rho050	frozenlake	0	Optimistic POMD (softmax)	0	2048	2048	2048
pathology_rho050	frozenlake	0	IDAC (frequentist actor-critic)	1	2048	128	n/a
pathology_rho050	frozenlake	0	UCB Q-learning (Jin et al.)	1	2048	32768	n/a
pathology_rho050	frozenlake	0	Optimistic POMD (softmax)	1	2048	2048	2048
pathology_rho050	frozenlake	0	IDAC (frequentist actor-critic)	2	2048	128	n/a
pathology_rho050	frozenlake	0	UCB Q-learning (Jin et al.)	2	2048	32768	n/a
pathology_rho050	frozenlake	0	Optimistic POMD (softmax)	2	2048	2048	2048
pathology_rho050	frozenlake	1	IDAC (frequentist actor-critic)	0	2048	128	n/a

Setting	Environment	Mode	Method	Seed	Episodes	Actor / Q updates	Episodes with all POMD costs zero
pathology_rho050	frozenlake	1	UCB Q-learning (Jin et al.)	0	2048	32768	n/a
pathology_rho050	frozenlake	1	Optimistic POMD (softmax)	0	2048	2048	2048
pathology_rho050	frozenlake	1	IDAC (frequentist actor-critic)	1	2048	128	n/a
pathology_rho050	frozenlake	1	UCB Q-learning (Jin et al.)	1	2048	32768	n/a
pathology_rho050	frozenlake	1	Optimistic POMD (softmax)	1	2048	2048	2048
pathology_rho050	frozenlake	1	IDAC (frequentist actor-critic)	2	2048	128	n/a
pathology_rho050	frozenlake	1	UCB Q-learning (Jin et al.)	2	2048	32768	n/a
pathology_rho050	frozenlake	1	Optimistic POMD (softmax)	2	2048	2048	2048
pathology_rho075	deepsea	0	IDAC (frequentist actor-critic)	0	2048	128	n/a
pathology_rho075	deepsea	0	UCB Q-learning (Jin et al.)	0	2048	18432	n/a
pathology_rho075	deepsea	0	Optimistic POMD (softmax)	0	2048	2048	2048
pathology_rho075	deepsea	0	IDAC (frequentist actor-critic)	1	2048	128	n/a
pathology_rho075	deepsea	0	UCB Q-learning (Jin et al.)	1	2048	18432	n/a
pathology_rho075	deepsea	0	Optimistic POMD (softmax)	1	2048	2048	2048

Setting	Environment	Mode	Method	Seed	Episodes	Actor / Q updates	Episodes with all POMD costs zero
pathology_rho075	deepsea	0	IDAC (frequentist actor-critic)	2	2048	128	n/a
pathology_rho075	deepsea	0	UCB Q-learning (Jin et al.)	2	2048	18432	n/a
pathology_rho075	deepsea	0	Optimistic POMD (softmax)	2	2048	2048	2048
pathology_rho075	deepsea	1	IDAC (frequentist actor-critic)	0	2048	128	n/a
pathology_rho075	deepsea	1	UCB Q-learning (Jin et al.)	0	2048	18432	n/a
pathology_rho075	deepsea	1	Optimistic POMD (softmax)	0	2048	2048	2048
pathology_rho075	deepsea	1	IDAC (frequentist actor-critic)	1	2048	128	n/a
pathology_rho075	deepsea	1	UCB Q-learning (Jin et al.)	1	2048	18432	n/a
pathology_rho075	deepsea	1	Optimistic POMD (softmax)	1	2048	2048	2048
pathology_rho075	deepsea	1	IDAC (frequentist actor-critic)	2	2048	128	n/a
pathology_rho075	deepsea	1	UCB Q-learning (Jin et al.)	2	2048	18432	n/a
pathology_rho075	deepsea	1	Optimistic POMD (softmax)	2	2048	2048	2048
pathology_rho075	frozenlake	0	IDAC (frequentist actor-critic)	0	2048	128	n/a
pathology_rho075	frozenlake	0	UCB Q-learning (Jin et al.)	0	2048	32768	n/a

Setting	Environment	Mode	Method	Seed	Episodes	Actor / Q updates	Episodes with all POMD costs zero
pathology_rho075	frozenlake	0	Optimistic POMD (softmax)	0	2048	2048	2048
pathology_rho075	frozenlake	0	IDAC (frequentist actor-critic)	1	2048	128	n/a
pathology_rho075	frozenlake	0	UCB Q-learning (Jin et al.)	1	2048	32768	n/a
pathology_rho075	frozenlake	0	Optimistic POMD (softmax)	1	2048	2048	2048
pathology_rho075	frozenlake	0	IDAC (frequentist actor-critic)	2	2048	128	n/a
pathology_rho075	frozenlake	0	UCB Q-learning (Jin et al.)	2	2048	32768	n/a
pathology_rho075	frozenlake	0	Optimistic POMD (softmax)	2	2048	2048	2048
pathology_rho075	frozenlake	1	IDAC (frequentist actor-critic)	0	2048	128	n/a
pathology_rho075	frozenlake	1	UCB Q-learning (Jin et al.)	0	2048	32768	n/a
pathology_rho075	frozenlake	1	Optimistic POMD (softmax)	0	2048	2048	2048
pathology_rho075	frozenlake	1	IDAC (frequentist actor-critic)	1	2048	128	n/a
pathology_rho075	frozenlake	1	UCB Q-learning (Jin et al.)	1	2048	32768	n/a
pathology_rho075	frozenlake	1	Optimistic POMD (softmax)	1	2048	2048	2048
pathology_rho075	frozenlake	1	IDAC (frequentist actor-critic)	2	2048	128	n/a

Setting	Environment	Mode	Method	Seed	Episodes	Actor / Q updates	Episodes with all POMD costs zero
pathology_rho075	frozenlake	1	UCB Q-learning (Jin et al.)	2	2048	32768	n/a
pathology_rho075	frozenlake	1	Optimistic POMD (softmax)	2	2048	2048	2048

IDAC updates both its cost critic and shadow critic at each reported actor update. UCB-Q counts updates on the analytically padded horizon; physical simulator steps remain separately charged. All three controls use the public finite domain and reward envelope, while CEAC/IDS additionally use declared candidate-model laws. That information-access difference must accompany any ranking.

Comparator provenance

Excluded 0 historical custom-control runs from these comparisons. Raw run hashes and source archives remain unchanged. Active primitive-action controls are IDAC (the upstream tabular two-critic/softmax update), UCB Q-learning (Jin et al., 2018, Algorithm 1), and optimistic POMD with a softmax policy (Shani et al., 2020, Algorithm 2). Their exact source, update equations, reward normalization and reset/padding adapter are documented in the comparator ledger. No cached policy-mixture OIR or policy-bank UCB curve is relabeled as a new algorithm.

Time traces

Expected policy return is evaluated exactly in the realized candidate model **by the evaluator only**. Pseudo-regret sums its gap to the model-optimal policy. Expected model KL measures decision-relevant evidence, not raw state coverage. The return trace uses a causal 16-episode moving mean within each seed before aggregation; other traces are cumulative. Observed log-likelihood-ratio traces are reconstructed from saved trajectories and oriented toward the realized mode for evaluation only. They differ from expected model KL, which is an exact evaluator quantity.

Lines: seed means. Bands: mean ±1 sample SD across seeds, not a confidence interval. No band for a single seed. There are 3 seeds per curve in this report. The episode index is not treated as additional independent replication. These are pointwise descriptive bands, not confidence intervals or simultaneous coverage guarantees. Nonnegative plot domains may clip a negative mean-minus-SD edge; exported values are untruncated.

The compressed `trace_statistics.csv.gz` records seed count, mean, sample SD, range, band edges and smoothing for every exported time point.

Endpoints

Setting	Environment	Context	Mode	Method	Seeds	Last-window expected return	Cumulative pseudo-regret	Primitive steps
pathology_rho075	deepsea	0	0	CEAC (exact-model single loop)	3	0.484000	0.477790	16394
pathology_rho075	deepsea	0	0	IDAC (frequentist actor-critic)	3	-0.014474	1020.492624	16547

Setting	Environment	Context	Mode	Method	Seeds	Last-window expected return	Cumulative pseudo-regret	Primitive steps
pathology_rho0.05	deepsea	0	0	Bayesian IDS: full support	3	0.484000	0.121667	16389
pathology_rho0.05	deepsea	0	0	Bayesian IDS: misspecified prior	3	0.484000	0.000000	16384
pathology_rho0.05	deepsea	0	0	Offline Graves–Lai oracle	3	0.484000	0.200000	16394
pathology_rho0.05	deepsea	0	0	Optimistic POMD (softmax)	3	-0.007526	1006.644722	16565
pathology_rho0.05	deepsea	0	0	UCB Q-learning (Jin et al.)	3	-0.023278	1039.457457	16503
pathology_rho0.05	deepsea	0	1	CEAC (exact-model single loop)	3	0.534000	0.641052	16388
pathology_rho0.05	deepsea	0	1	IDAC (frequentist actor-critic)	3	-0.001952	1097.053113	16544
pathology_rho0.05	deepsea	0	1	Bayesian IDS: full support	3	0.534000	0.088333	16388
pathology_rho0.05	deepsea	0	1	Bayesian IDS: misspecified prior	3	0.484000	102.400000	16384
pathology_rho0.05	deepsea	0	1	Offline Graves–Lai oracle	3	0.534000	0.000000	16384
pathology_rho0.05	deepsea	0	1	Optimistic POMD (softmax)	3	0.003037	1087.412722	16565
pathology_rho0.05	deepsea	0	1	UCB Q-learning (Jin et al.)	3	0.006357	1082.236974	16503
pathology_rho0.05	frozenlake	0	0	CEAC (exact-model single loop)	3	0.489000	1.113550	22546

Setting	Environment	Context	Mode	Method	Seeds	Last-window expected return	Cumulative pseudo-regret	Primitive steps
pathology_rho0.5	frozenlake	0	0	IDAC (frequentist actor-critic)	3	-0.025171	1053.017869	32739
pathology_rho0.5	frozenlake	0	0	Bayesian IDS: full support	3	0.489000	0.147000	22540
pathology_rho0.5	frozenlake	0	0	Bayesian IDS: misspecified prior	3	0.489000	0.000000	22528
pathology_rho0.5	frozenlake	0	0	Offline Graves–Lai oracle	3	0.489000	0.220000	22548
pathology_rho0.5	frozenlake	0	0	Optimistic POMD (softmax)	3	-0.025057	1052.789621	32734
pathology_rho0.5	frozenlake	0	0	UCB Q-learning (Jin et al.)	3	-0.033500	1068.692792	32768
pathology_rho0.5	frozenlake	0	1	CEAC (exact-model single loop)	3	0.539000	1.554215	26611
pathology_rho0.5	frozenlake	0	1	IDAC (frequentist actor-critic)	3	-0.025343	1155.760338	32752
pathology_rho0.5	frozenlake	0	1	Bayesian IDS: full support	3	0.539000	0.063833	26629
pathology_rho0.5	frozenlake	0	1	Bayesian IDS: misspecified prior	3	0.489000	102.400000	22528
pathology_rho0.5	frozenlake	0	1	Offline Graves–Lai oracle	3	0.539000	0.000000	26624
pathology_rho0.5	frozenlake	0	1	Optimistic POMD (softmax)	3	-0.025047	1155.167331	32734
pathology_rho0.5	frozenlake	0	1	UCB Q-learning (Jin et al.)	3	-0.033500	1171.091333	32768

Setting	Environment	Context	Mode	Method	Seeds	Last-window expected return	Cumulative pseudo-regret	Primitive steps
pathology_rho025	deepsea	0	0	CEAC (exact-model single loop)	3	0.484000	0.859444	16394
pathology_rho025	deepsea	0	0	IDAC (frequentist actor-critic)	3	-0.020872	1033.871737	16508
pathology_rho025	deepsea	0	0	Bayesian IDS: full support	3	0.484000	0.121667	16389
pathology_rho025	deepsea	0	0	Bayesian IDS: misspecified prior	3	0.484000	0.000000	16384
pathology_rho025	deepsea	0	0	Offline Graves–Lai oracle	3	0.484000	0.200000	16394
pathology_rho025	deepsea	0	0	Optimistic POMD (softmax)	3	-0.016352	1024.721490	16530
pathology_rho025	deepsea	0	0	UCB Q-learning (Jin et al.)	3	-0.030809	1054.218030	16437
pathology_rho025	deepsea	0	1	CEAC (exact-model single loop)	3	0.534000	1.035606	16388
pathology_rho025	deepsea	0	1	IDAC (frequentist actor-critic)	3	-0.011454	1116.801680	16513
pathology_rho025	deepsea	0	1	Bayesian IDS: full support	3	0.534000	0.095000	16388
pathology_rho025	deepsea	0	1	Bayesian IDS: misspecified prior	3	0.484000	102.400000	16384
pathology_rho025	deepsea	0	1	Offline Graves–Lai oracle	3	0.534000	0.000000	16384
pathology_rho025	deepsea	0	1	Optimistic POMD (softmax)	3	-0.007845	1109.699268	16530

Setting	Environment	Context	Mode	Method	Seeds	Last-window expected return	Cumulative pseudo-regret	Primitive steps
pathology_rho025	deepsea	0	1	UCB Q-learning (Jin et al.)	3	-0.016613	1127.179505	16437
pathology_rho025	frozenlake	0	0	CEAC (exact-model single loop)	3	0.489000	1.113750	22546
pathology_rho025	frozenlake	0	0	IDAC (frequentist actor-critic)	3	-0.025175	1053.023342	32741
pathology_rho025	frozenlake	0	0	Bayesian IDS: full support	3	0.489000	0.147000	22540
pathology_rho025	frozenlake	0	0	Bayesian IDS: misspecified prior	3	0.489000	0.000000	22528
pathology_rho025	frozenlake	0	0	Offline Graves–Lai oracle	3	0.489000	0.220000	22548
pathology_rho025	frozenlake	0	0	Optimistic POMD (softmax)	3	-0.025066	1052.807275	32737
pathology_rho025	frozenlake	0	0	UCB Q-learning (Jin et al.)	3	-0.034333	1070.391115	32768
pathology_rho025	frozenlake	0	1	CEAC (exact-model single loop)	3	0.539000	1.554446	26611
pathology_rho025	frozenlake	0	1	IDAC (frequentist actor-critic)	3	-0.025338	1155.749622	32754
pathology_rho025	frozenlake	0	1	Bayesian IDS: full support	3	0.539000	0.063833	26629
pathology_rho025	frozenlake	0	1	Bayesian IDS: misspecified prior	3	0.489000	102.400000	22528
pathology_rho025	frozenlake	0	1	Offline Graves–Lai oracle	3	0.539000	0.000000	26624

Setting	Environment	Context	Mode	Method	Seeds	Last-window expected return	Cumulative pseudo-regret	Primitive steps
pathology_rho025	frozenlake	0	1	Optimistic POMD (softmax)	3	-0.025055	1155.184985	32737
pathology_rho025	frozenlake	0	1	UCB Q-learning (Jin et al.)	3	-0.034333	1172.786948	32768
pathology_rho050	deepsea	0	0	CEAC (exact-model single loop)	3	0.484000	1.032547	16394
pathology_rho050	deepsea	0	0	IDAC (frequentist actor-critic)	3	-0.022988	1038.372445	16465
pathology_rho050	deepsea	0	0	Bayesian IDS: full support	3	0.484000	0.115000	16389
pathology_rho050	deepsea	0	0	Bayesian IDS: misspecified prior	3	0.484000	0.000000	16384
pathology_rho050	deepsea	0	0	Offline Graves–Lai oracle	3	0.484000	0.200000	16394
pathology_rho050	deepsea	0	0	Optimistic POMD (softmax)	3	-0.020329	1032.866667	16481
pathology_rho050	deepsea	0	0	UCB Q-learning (Jin et al.)	3	-0.021979	1036.512911	16394
pathology_rho050	deepsea	0	1	CEAC (exact-model single loop)	3	0.534000	1.223996	16388
pathology_rho050	deepsea	0	1	IDAC (frequentist actor-critic)	3	-0.016913	1128.127073	16465
pathology_rho050	deepsea	0	1	Bayesian IDS: full support	3	0.534000	0.095000	16388
pathology_rho050	deepsea	0	1	Bayesian IDS: misspecified prior	3	0.484000	102.400000	16384

Setting	Environment	Context	Mode	Method	Seeds	Last-window expected return	Cumulative pseudo-regret	Primitive steps
pathology_rho0.50	deepsea	0	1	Offline Graves–Lai oracle	3	0.534000	0.000000	16384
pathology_rho0.50	deepsea	0	1	Optimistic POMD (softmax)	3	-0.014079	1122.466667	16481
pathology_rho0.50	deepsea	0	1	UCB Q-learning (Jin et al.)	3	-0.010734	1116.034953	16394
pathology_rho0.50	frozenlake	0	0	CEAC (exact-model single loop)	3	0.489000	1.114000	22546
pathology_rho0.50	frozenlake	0	0	IDAC (frequentist actor-critic)	3	-0.025172	1053.019479	32739
pathology_rho0.50	frozenlake	0	0	Bayesian IDS: full support	3	0.489000	0.147000	22540
pathology_rho0.50	frozenlake	0	0	Bayesian IDS: misspecified prior	3	0.489000	0.000000	22528
pathology_rho0.50	frozenlake	0	0	Offline Graves–Lai oracle	3	0.489000	0.220000	22548
pathology_rho0.50	frozenlake	0	0	Optimistic POMD (softmax)	3	-0.025077	1052.829343	32735
pathology_rho0.50	frozenlake	0	0	UCB Q-learning (Jin et al.)	3	-0.033917	1069.529583	32768
pathology_rho0.50	frozenlake	0	1	CEAC (exact-model single loop)	3	0.539000	1.554735	26611
pathology_rho0.50	frozenlake	0	1	IDAC (frequentist actor-critic)	3	-0.025325	1155.722262	32749
pathology_rho0.50	frozenlake	0	1	Bayesian IDS: full support	3	0.539000	0.063833	26629

Setting	Environment	Context	Mode	Method	Seeds	Last-window expected return	Cumulative pseudo-regret	Primitive steps
pathology_rho050	frozenlake	0	1	Bayesian IDS: misspecified prior	3	0.489000	102.400000	22528
pathology_rho050	frozenlake	0	1	Offline Graves–Lai oracle	3	0.539000	0.000000	26624
pathology_rho050	frozenlake	0	1	Optimistic POMD (softmax)	3	-0.025066	1155.207053	32735
pathology_rho050	frozenlake	0	1	UCB Q-learning (Jin et al.)	3	-0.033917	1171.927500	32768
pathology_rho075	deepsea	0	0	CEAC (exact-model single loop)	3	0.484000	1.159492	16394
pathology_rho075	deepsea	0	0	IDAC (frequentist actor-critic)	3	-0.022552	1037.500059	16425
pathology_rho075	deepsea	0	0	Bayesian IDS: full support	3	0.484000	0.121667	16389
pathology_rho075	deepsea	0	0	Bayesian IDS: misspecified prior	3	0.484000	0.000000	16384
pathology_rho075	deepsea	0	0	Offline Graves–Lai oracle	3	0.484000	0.200000	16394
pathology_rho075	deepsea	0	0	Optimistic POMD (softmax)	3	-0.020904	1034.042503	16463
pathology_rho075	deepsea	0	0	UCB Q-learning (Jin et al.)	3	-0.020363	1033.429520	16386
pathology_rho075	deepsea	0	1	CEAC (exact-model single loop)	3	0.534000	1.365359	16388
pathology_rho075	deepsea	0	1	IDAC (frequentist actor-critic)	3	-0.019254	1132.989785	16425

Setting	Environment	Context	Mode	Method	Seeds	Last-window expected return	Cumulative pseudo-regret	Primitive steps
pathology_rho075	deepsea	0	1	Bayesian IDS: full support	3	0.534000	0.088333	16388
pathology_rho075	deepsea	0	1	Bayesian IDS: misspecified prior	3	0.484000	102.400000	16384
pathology_rho075	deepsea	0	1	Offline Graves–Lai oracle	3	0.534000	0.000000	16384
pathology_rho075	deepsea	0	1	Optimistic POMD (softmax)	3	-0.016563	1127.553614	16463
pathology_rho075	deepsea	0	1	UCB Q-learning (Jin et al.)	3	-0.018016	1130.977433	16386
pathology_rho075	frozenlake	0	0	CEAC (exact-model single loop)	3	0.489000	1.114250	22546
pathology_rho075	frozenlake	0	0	IDAC (frequentist actor-critic)	3	-0.025171	1053.016226	32738
pathology_rho075	frozenlake	0	0	Bayesian IDS: full support	3	0.489000	0.147000	22540
pathology_rho075	frozenlake	0	0	Bayesian IDS: misspecified prior	3	0.489000	0.000000	22528
pathology_rho075	frozenlake	0	0	Offline Graves–Lai oracle	3	0.489000	0.220000	22548
pathology_rho075	frozenlake	0	0	Optimistic POMD (softmax)	3	-0.025088	1052.851411	32736
pathology_rho075	frozenlake	0	0	UCB Q-learning (Jin et al.)	3	-0.034542	1070.812740	32768
pathology_rho075	frozenlake	0	1	CEAC (exact-model single loop)	3	0.539000	1.555024	26611

Setting	Environment	Context	Mode	Method	Seeds	Last-window expected return	Cumulative pseudo-regret	Primitive steps
pathology_rho075	frozenlake	0	1	IDAC (frequentist actor-critic)	3	-0.025316	1155.706615	32753
pathology_rho075	frozenlake	0	1	Bayesian IDS: full support	3	0.539000	0.063833	26629
pathology_rho075	frozenlake	0	1	Bayesian IDS: misspecified prior	3	0.489000	102.400000	22528
pathology_rho075	frozenlake	0	1	Offline Graves–Lai oracle	3	0.539000	0.000000	26624
pathology_rho075	frozenlake	0	1	Optimistic POMD (softmax)	3	-0.025077	1155.229121	32736
pathology_rho075	frozenlake	0	1	UCB Q-learning (Jin et al.)	3	-0.034542	1173.211698	32768

Budgets are matched in episodes. Episode lengths differ; the table also reports primitive environment steps. Planning uses the declared finite models and is not charged as physical self-play. This is not a wall-clock-matched comparison. Terminal success is not interchangeable with return: a confounded route can reach a dock reliably while missing the better objective.

Deployment versus proposal actor

Setting	Environment	Mode	Method	Seed	Actor episodes	Final actor return	Certified response return	Deployment episodes
pathology_rho065	deepsea	0	CEAC (exact-model single loop)	0	0	0.365084	0.484000	2034
pathology_rho065	deepsea	0	CEAC (exact-model single loop)	1	0	0.365084	0.484000	2034
pathology_rho065	deepsea	0	CEAC (exact-model single loop)	2	0	0.365084	0.484000	2034
pathology_rho065	deepsea	1	CEAC (exact-model single loop)	0	0	0.367831	0.534000	1632

Setting	Environment	Mode	Method	Seed	Actor episodes	Final actor return	Certified response return	Deployment episodes
pathology_rho0.5	deepsea	1	CEAC (exact-model single loop)	1	2	0.367930	0.534000	1638
pathology_rho0.5	deepsea	1	CEAC (exact-model single loop)	2	0	0.367831	0.534000	1577
pathology_rho0.5	frozenlake	0	CEAC (exact-model single loop)	0	0	0.469396	0.489000	2032
pathology_rho0.5	frozenlake	0	CEAC (exact-model single loop)	1	0	0.469396	0.489000	2032
pathology_rho0.5	frozenlake	0	CEAC (exact-model single loop)	2	0	0.469396	0.489000	2032
pathology_rho0.5	frozenlake	1	CEAC (exact-model single loop)	0	1	0.471673	0.539000	1638
pathology_rho0.5	frozenlake	1	CEAC (exact-model single loop)	1	14	0.471775	0.539000	1700
pathology_rho0.5	frozenlake	1	CEAC (exact-model single loop)	2	0	0.471672	0.539000	1540
pathology_rho0.25	deepsea	0	CEAC (exact-model single loop)	0	0	0.345794	0.484000	2034
pathology_rho0.25	deepsea	0	CEAC (exact-model single loop)	1	0	0.345794	0.484000	2034
pathology_rho0.25	deepsea	0	CEAC (exact-model single loop)	2	0	0.345794	0.484000	2034
pathology_rho0.25	deepsea	1	CEAC (exact-model single loop)	0	0	0.348503	0.534000	1632
pathology_rho0.25	deepsea	1	CEAC (exact-model single loop)	1	2	0.348562	0.534000	1638
pathology_rho0.25	deepsea	1	CEAC (exact-model single loop)	2	0	0.348503	0.534000	1577

Setting	Environment	Mode	Method	Seed	Actor episodes	Final actor return	Certified response return	Deployment episodes
pathology_rho025	frozenlake	0	CEAC (exact-model single loop)	0	0	0.469389	0.489000	2032
pathology_rho025	frozenlake	0	CEAC (exact-model single loop)	1	0	0.469389	0.489000	2032
pathology_rho025	frozenlake	0	CEAC (exact-model single loop)	2	0	0.469389	0.489000	2032
pathology_rho025	frozenlake	1	CEAC (exact-model single loop)	0	1	0.471667	0.539000	1638
pathology_rho025	frozenlake	1	CEAC (exact-model single loop)	1	14	0.471769	0.539000	1700
pathology_rho025	frozenlake	1	CEAC (exact-model single loop)	2	0	0.471666	0.539000	1540
pathology_rho050	deepsea	0	CEAC (exact-model single loop)	0	0	0.322910	0.484000	2034
pathology_rho050	deepsea	0	CEAC (exact-model single loop)	1	0	0.322910	0.484000	2034
pathology_rho050	deepsea	0	CEAC (exact-model single loop)	2	0	0.322910	0.484000	2034
pathology_rho050	deepsea	1	CEAC (exact-model single loop)	0	0	0.325573	0.534000	1632
pathology_rho050	deepsea	1	CEAC (exact-model single loop)	1	2	0.325641	0.534000	1638
pathology_rho050	deepsea	1	CEAC (exact-model single loop)	2	0	0.325573	0.534000	1577
pathology_rho050	frozenlake	0	CEAC (exact-model single loop)	0	0	0.469382	0.489000	2032
pathology_rho050	frozenlake	0	CEAC (exact-model single loop)	1	0	0.469382	0.489000	2032

Setting	Environment	Mode	Method	Seed	Actor episodes	Final actor return	Certified response return	Deployment episodes
pathology_rho050	frozenlake	0	CEAC (exact-model single loop)	2	0	0.469382	0.489000	2032
pathology_rho050	frozenlake	1	CEAC (exact-model single loop)	0	1	0.471659	0.539000	1638
pathology_rho050	frozenlake	1	CEAC (exact-model single loop)	1	14	0.471761	0.539000	1700
pathology_rho050	frozenlake	1	CEAC (exact-model single loop)	2	0	0.471658	0.539000	1540
pathology_rho075	deepsea	0	CEAC (exact-model single loop)	0	0	0.301330	0.484000	2034
pathology_rho075	deepsea	0	CEAC (exact-model single loop)	1	0	0.301330	0.484000	2034
pathology_rho075	deepsea	0	CEAC (exact-model single loop)	2	0	0.301330	0.484000	2034
pathology_rho075	deepsea	1	CEAC (exact-model single loop)	0	0	0.303946	0.534000	1632
pathology_rho075	deepsea	1	CEAC (exact-model single loop)	1	2	0.304023	0.534000	1638
pathology_rho075	deepsea	1	CEAC (exact-model single loop)	2	0	0.303946	0.534000	1577
pathology_rho075	frozenlake	0	CEAC (exact-model single loop)	0	0	0.469374	0.489000	2032
pathology_rho075	frozenlake	0	CEAC (exact-model single loop)	1	0	0.469374	0.489000	2032
pathology_rho075	frozenlake	0	CEAC (exact-model single loop)	2	0	0.469374	0.489000	2032
pathology_rho075	frozenlake	1	CEAC (exact-model single loop)	0	1	0.471651	0.539000	1638

Setting	Environment	Mode	Method	Seed	Actor episodes	Final actor return	Certified response return	Deployment episodes
pathology_rho075	frozenlake	1	CEAC (exact-model single loop)	1	14	0.471753	0.539000	1700
pathology_rho075	frozenlake	1	CEAC (exact-model single loop)	2	0	0.471650	0.539000	1540

Certified deployment trajectories remain policy-labeled evidence; they do not become off-policy actor-gradient updates. When included, the actor-only ablation shares initialization, design logic and random-stream identifiers, with histories diverging after policies differ.

Mathematical correction

Given a likelihood-confidence set $\mathcal{C}_t$, form a proposal from its confidence-restricted model estimate, then verify its reward gap in every plausible model:

$$\epsilon_t^{\text{deploy}} = \max_{m \in \mathcal{C}_t} \left[J_m^\star - J_m(\widehat{\pi}_t) \right].$$

CEAC permits certified deployment only when this residual is at most the declared numerical tolerance. On the simultaneous confidence event, the same bound holds for the true-model deployment gap. If the confidence set changes, the response and certificate are refreshed. No true-mode label is used to construct this policy.

Appendix D defines a deployment response; Section 4 and Appendix G require the reward-only certificate. The Section 3 selector did not explicitly connect that response to free-episode execution. The implementation makes that connection explicit, using exact finite-model planning. It is a documented execution refinement, not evidence that actor SGD suddenly converged.

Recorded 12396 CEAC/ablation refresh certificates. Maximum design primal/dual gap: 1.39e-17; maximum full-policy column residual: 2.22e-16.

Detailed specifications and remaining theorem-level limits: correction ledger. The v90 manuscript and previous experiment artifacts remain unchanged.

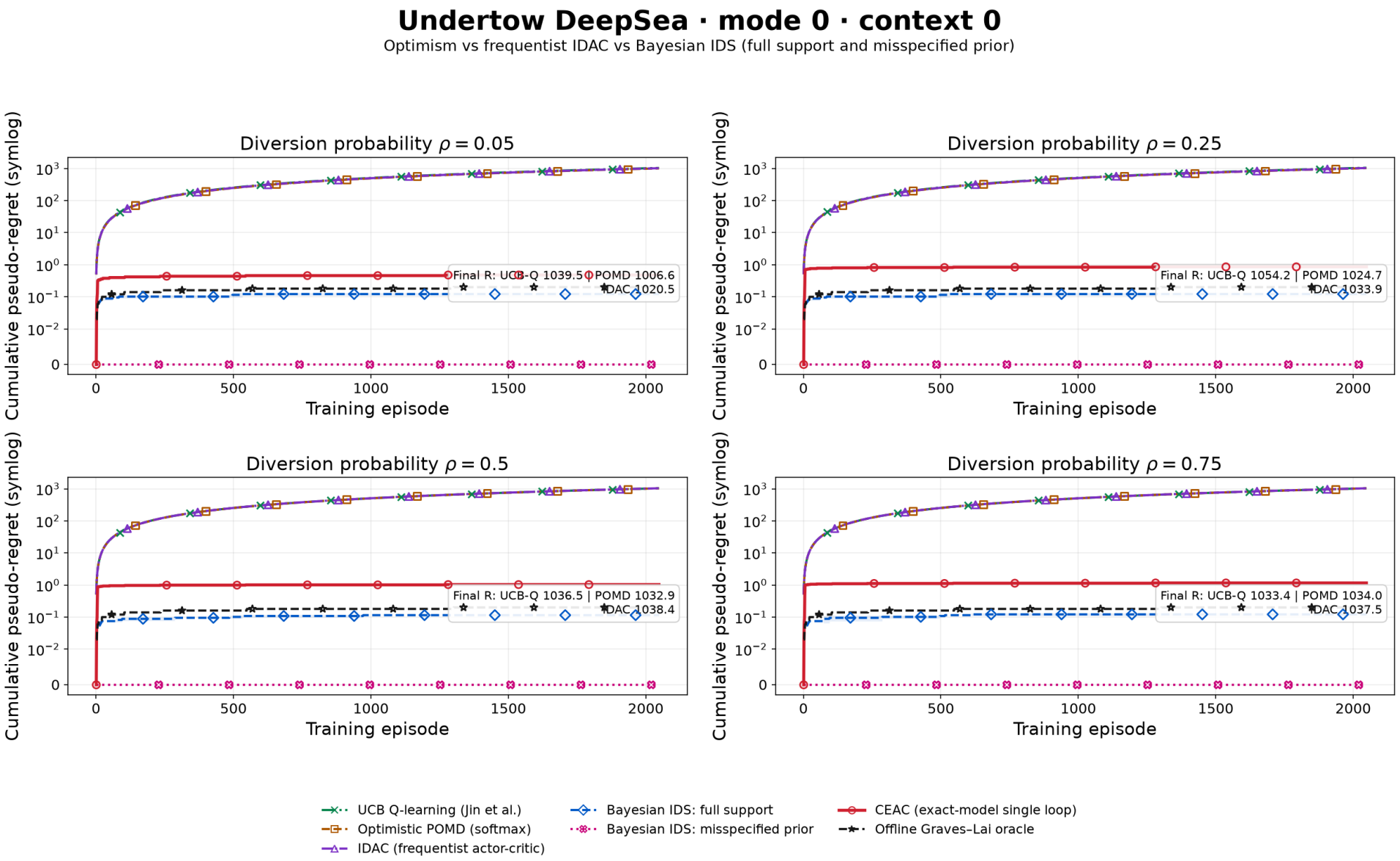


Figure 5: pathology overlay deepsea context0 mode0

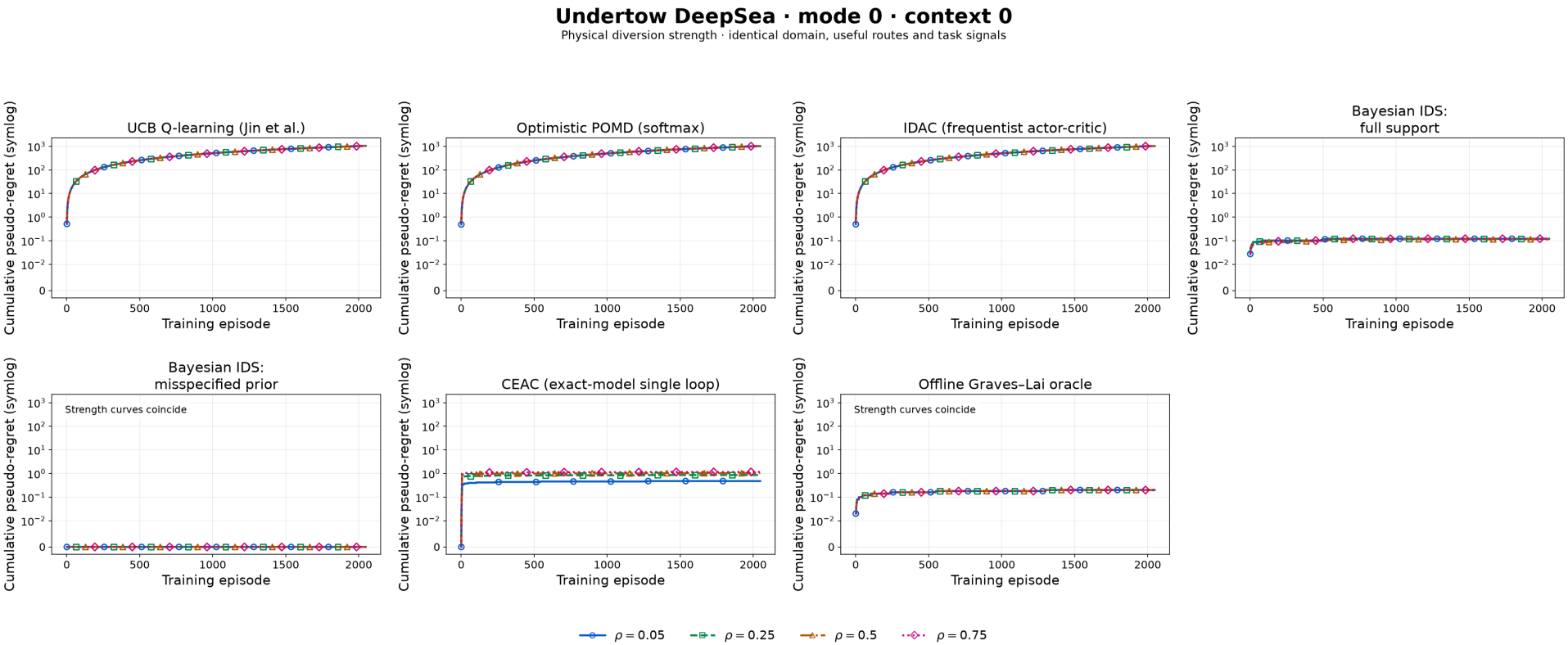


Figure 6: pathology learning deepsea context0 mode0

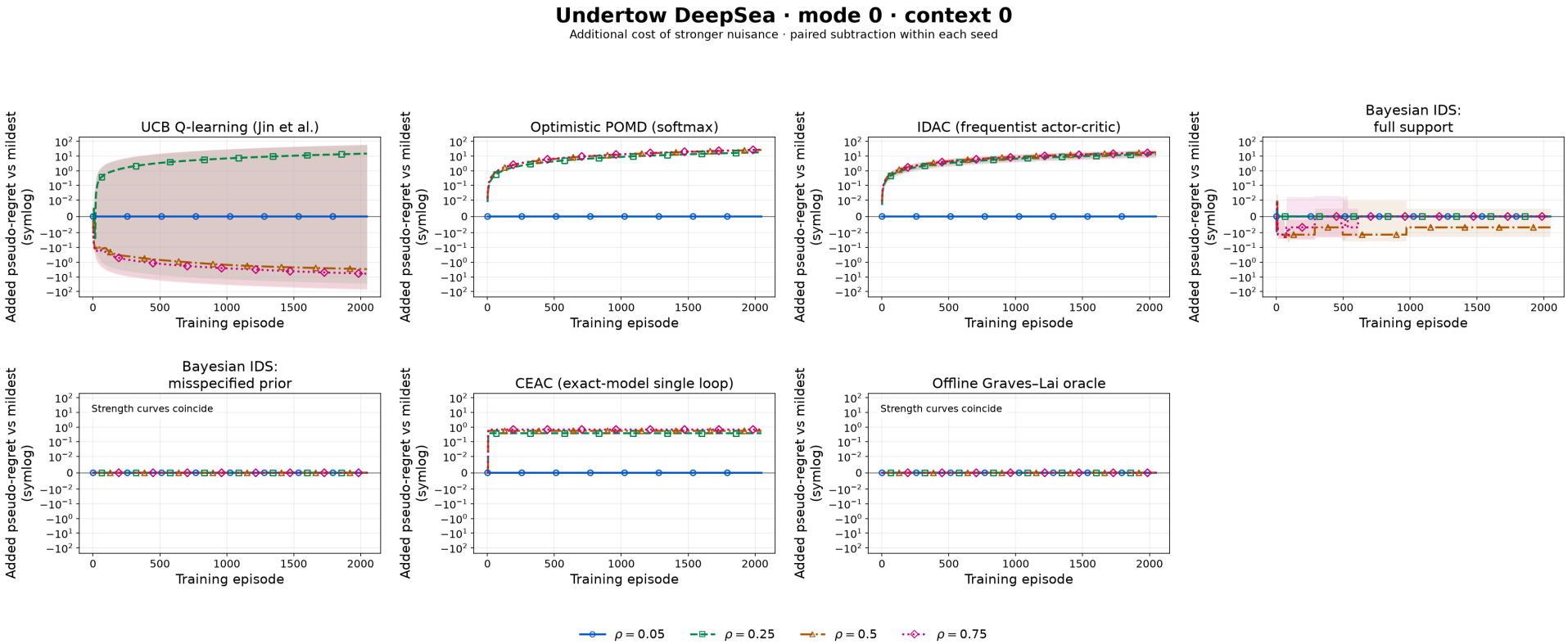


Figure 7: pathology added regret deepsea context0 mode0

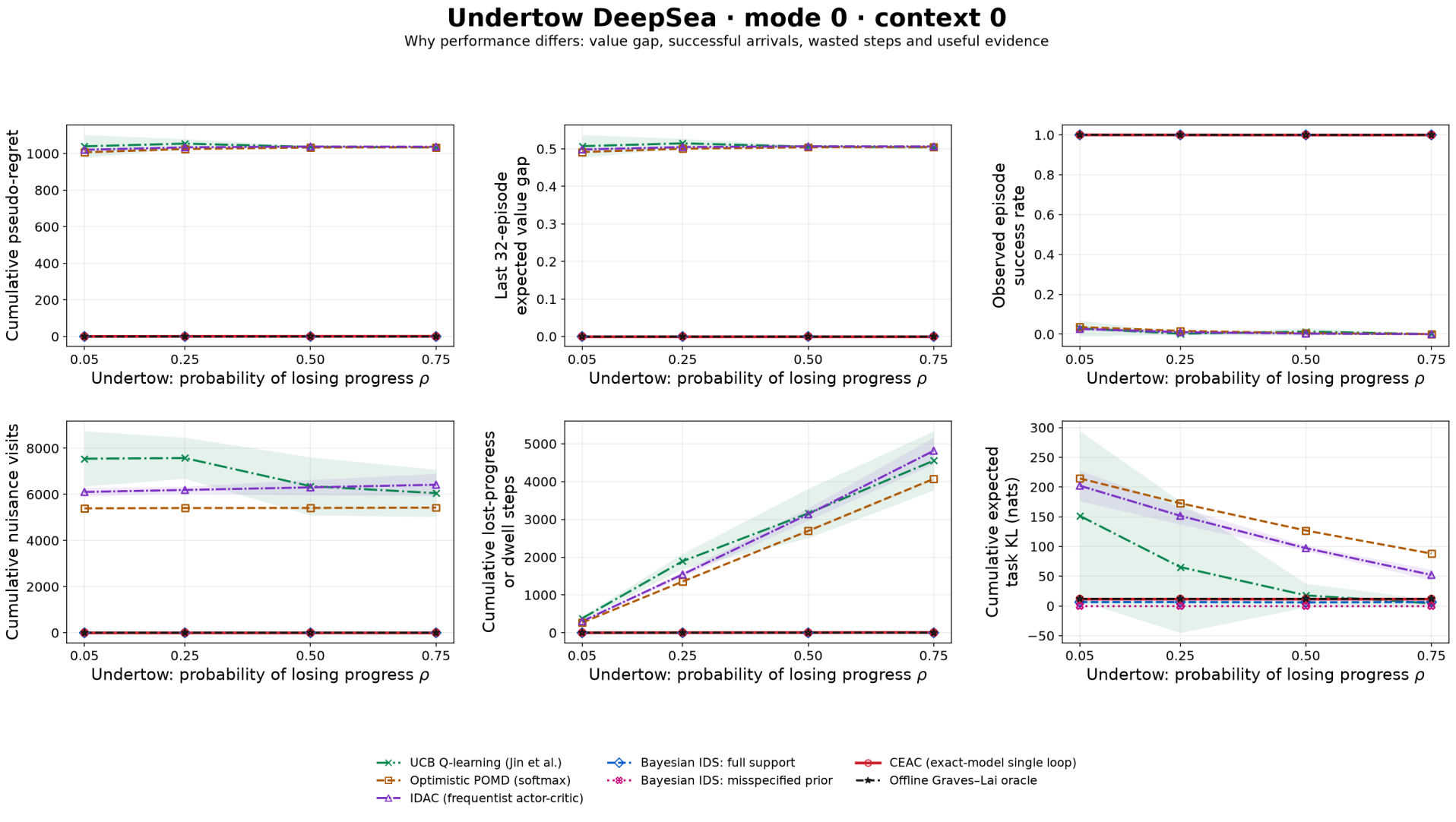


Figure 8: pathology diagnostics deepsea context0 mode0

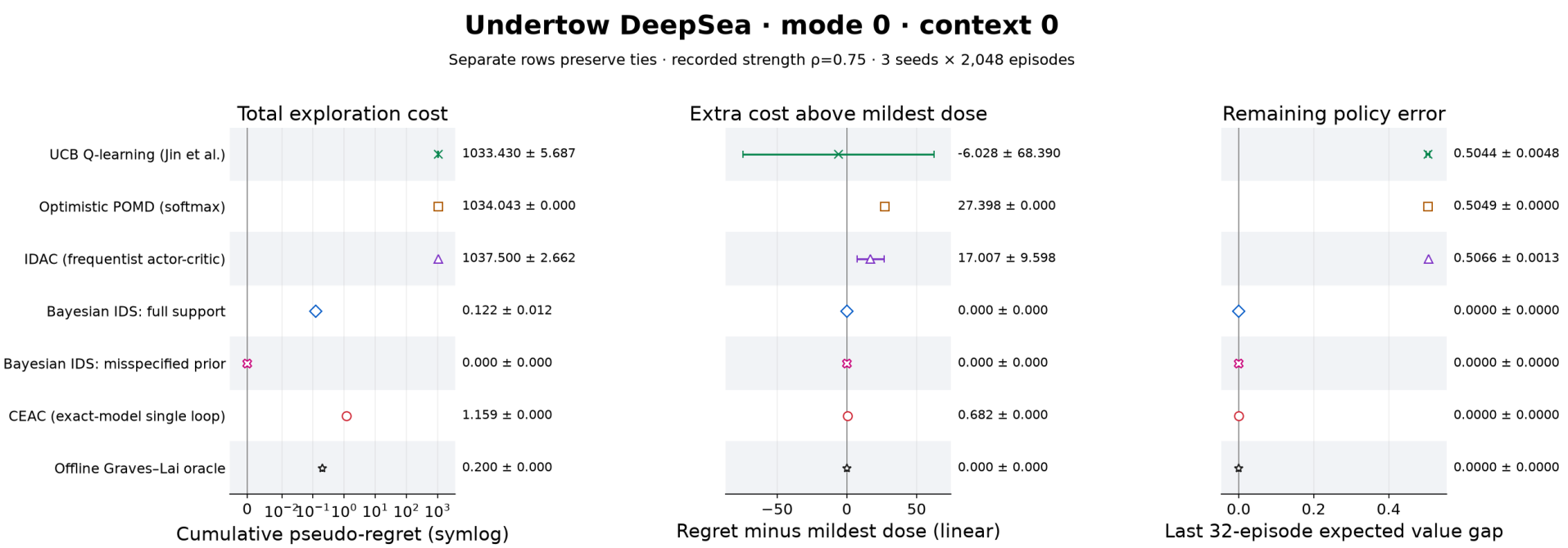


Figure 9: pathology endpoints deepsea context0 mode0

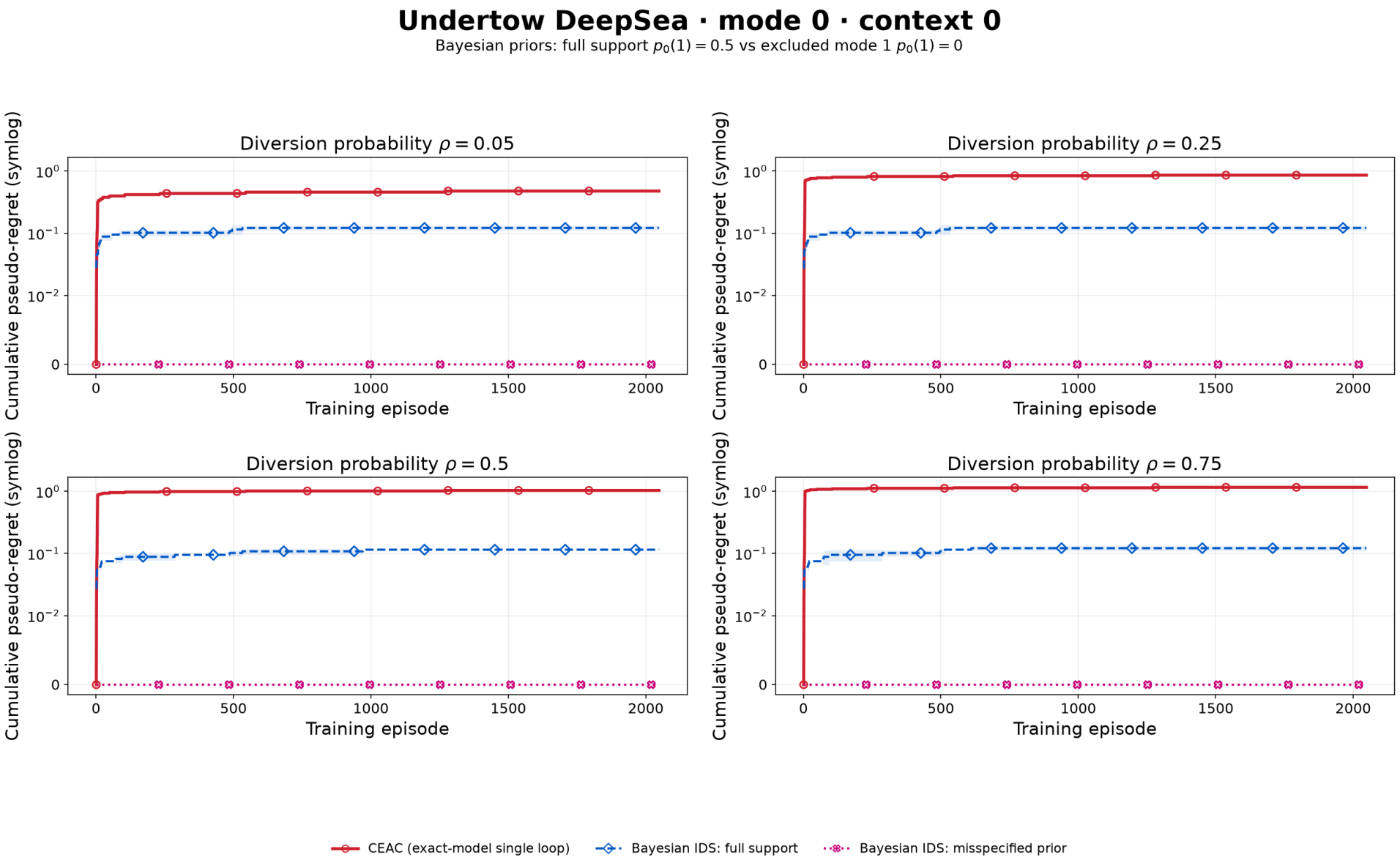


Figure 10: pathology prior deepsea context0 mode0

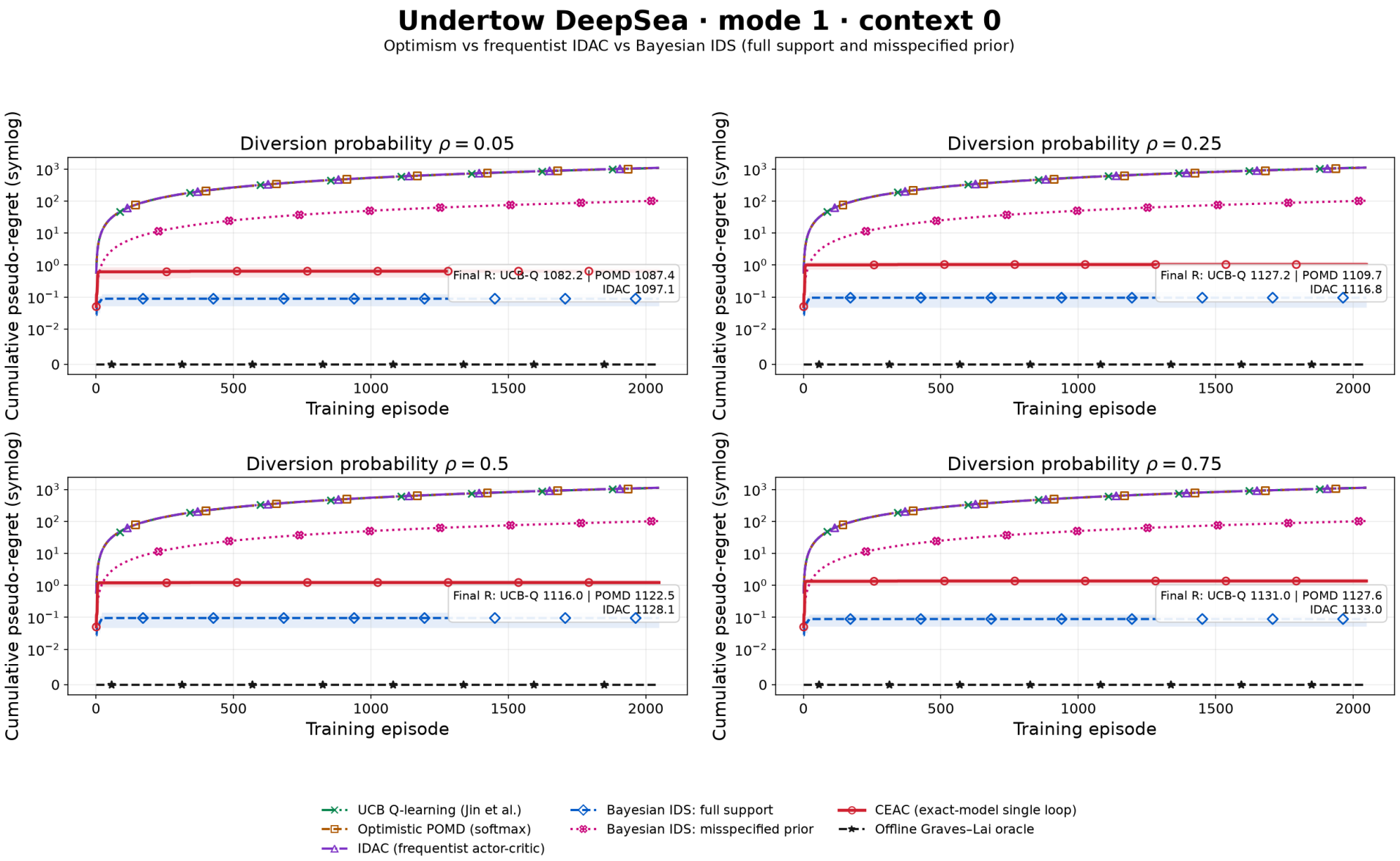


Figure 11: pathology overlay deepsea context0 mode1

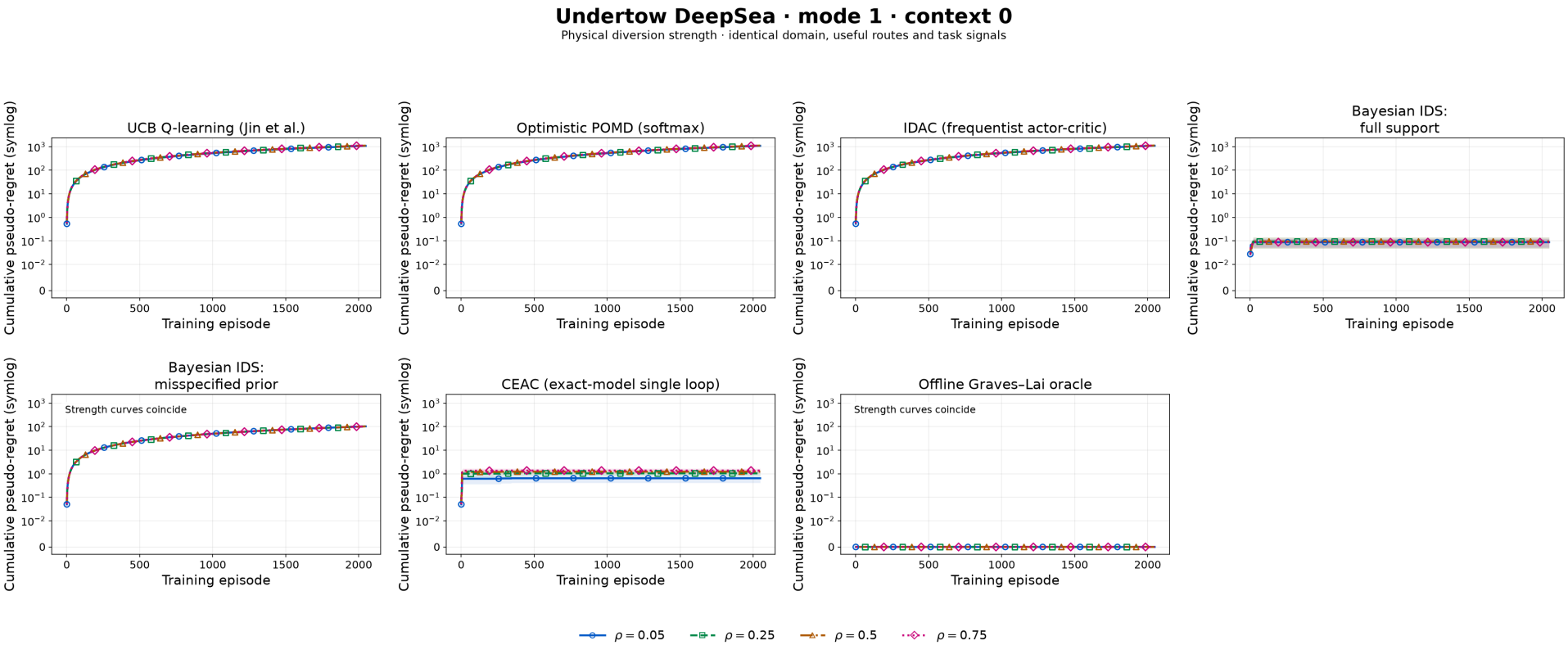


Figure 12: pathology learning deepsea context0 mode1

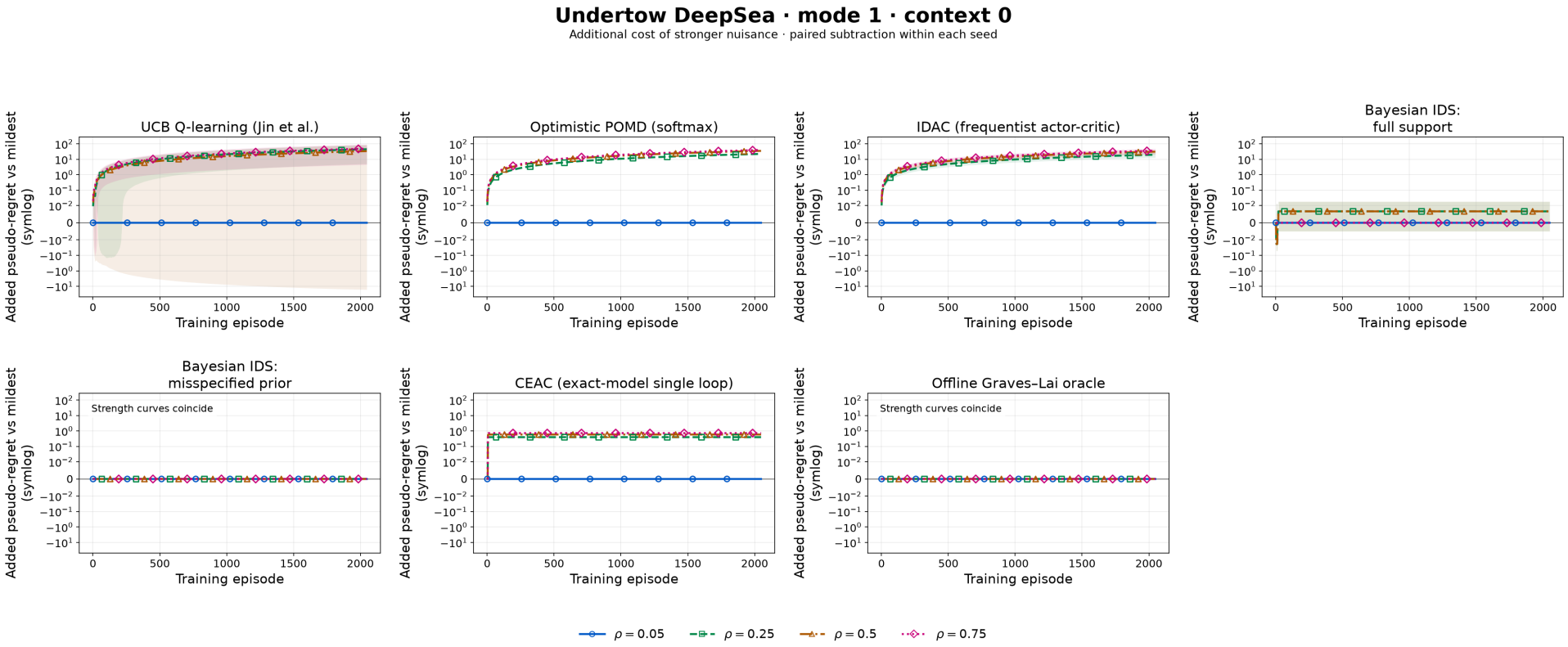


Figure 13: pathology added regret deepsea context0 mode1

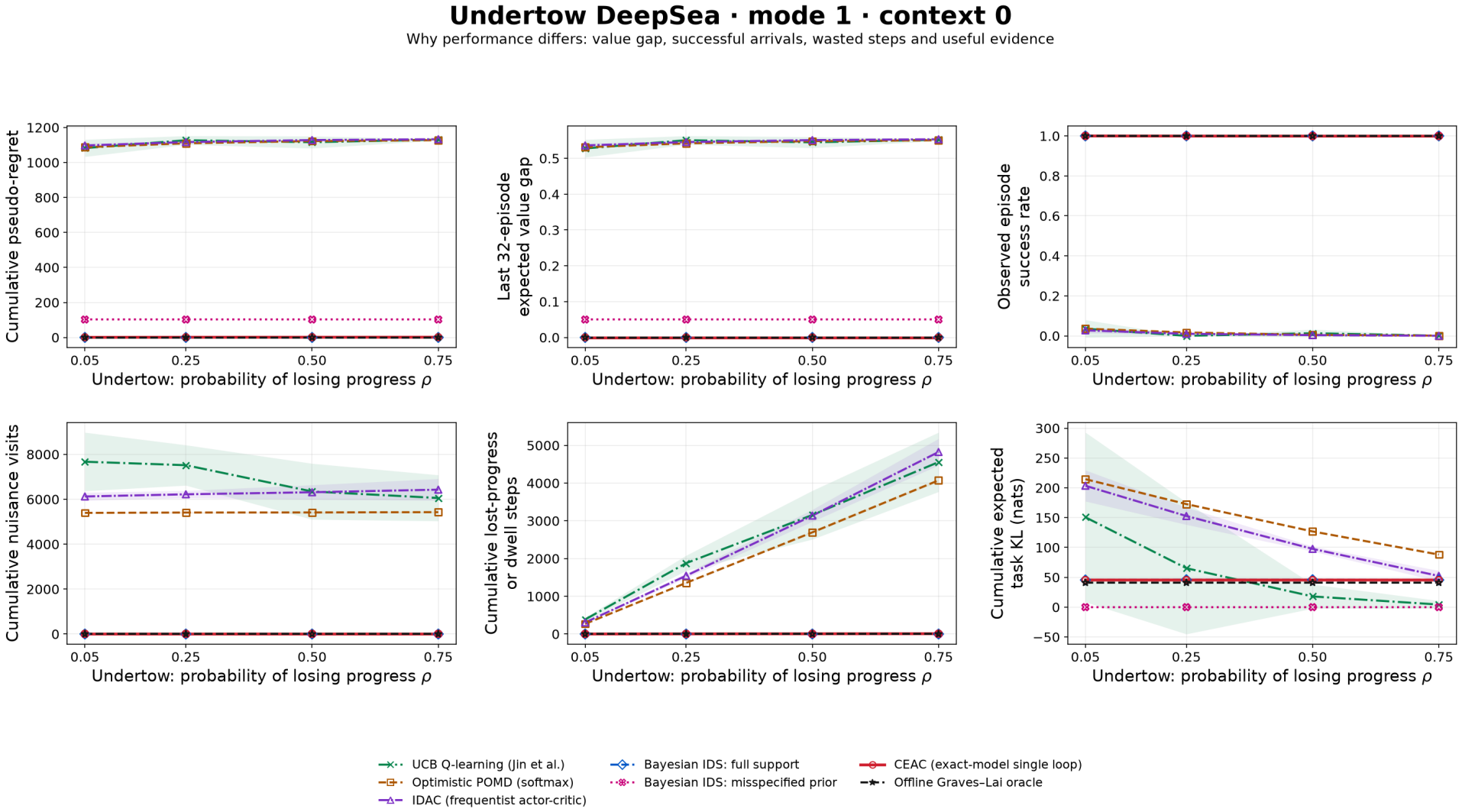


Figure 14: pathology diagnostics deepsea context0 mode1



Figure 15: pathology endpoints deepsea context0 mode1

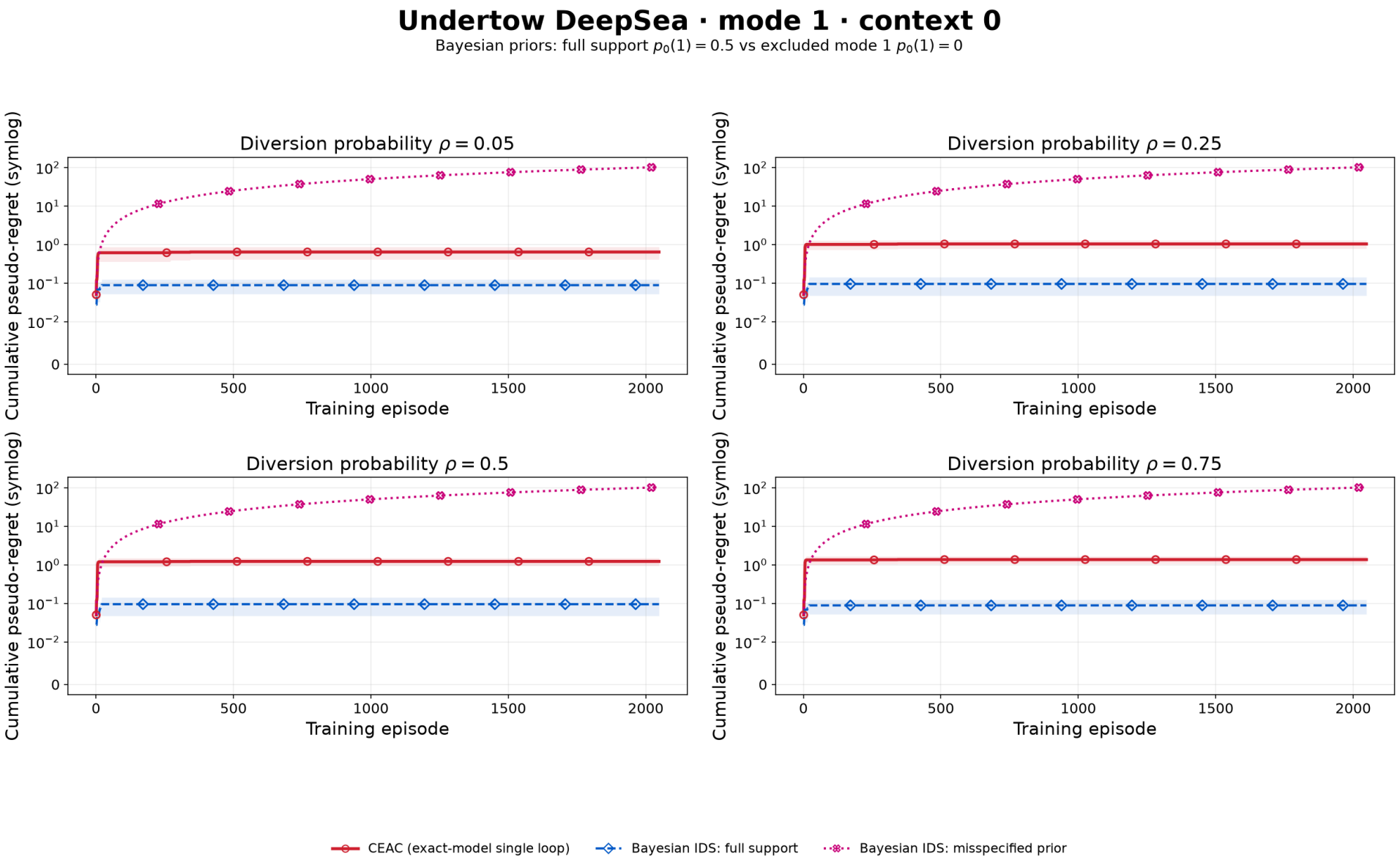


Figure 16: pathology prior deepsea context0 mode1

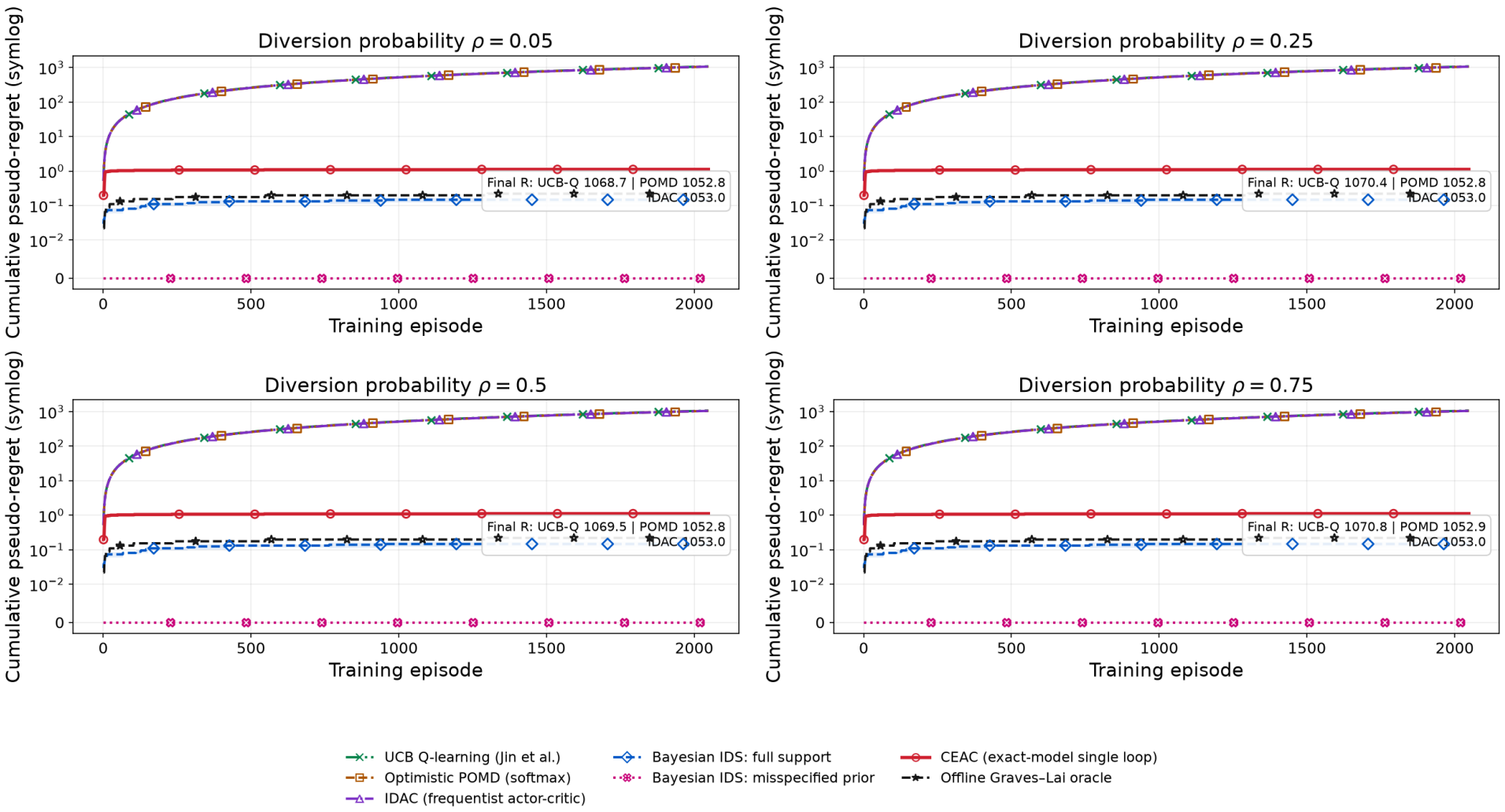


Figure 17: pathology overlay frozenlake context0 mode0

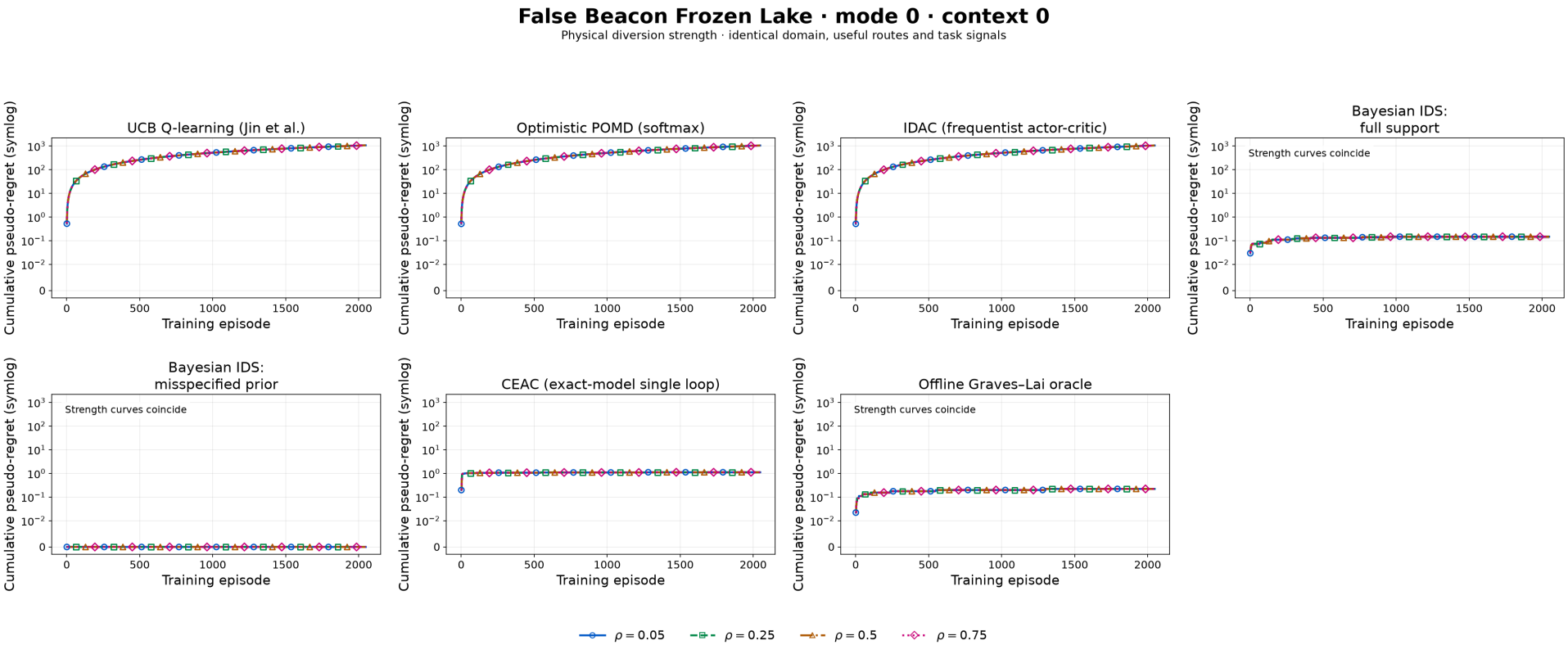


Figure 18: pathology learning frozenlake context0 mode0

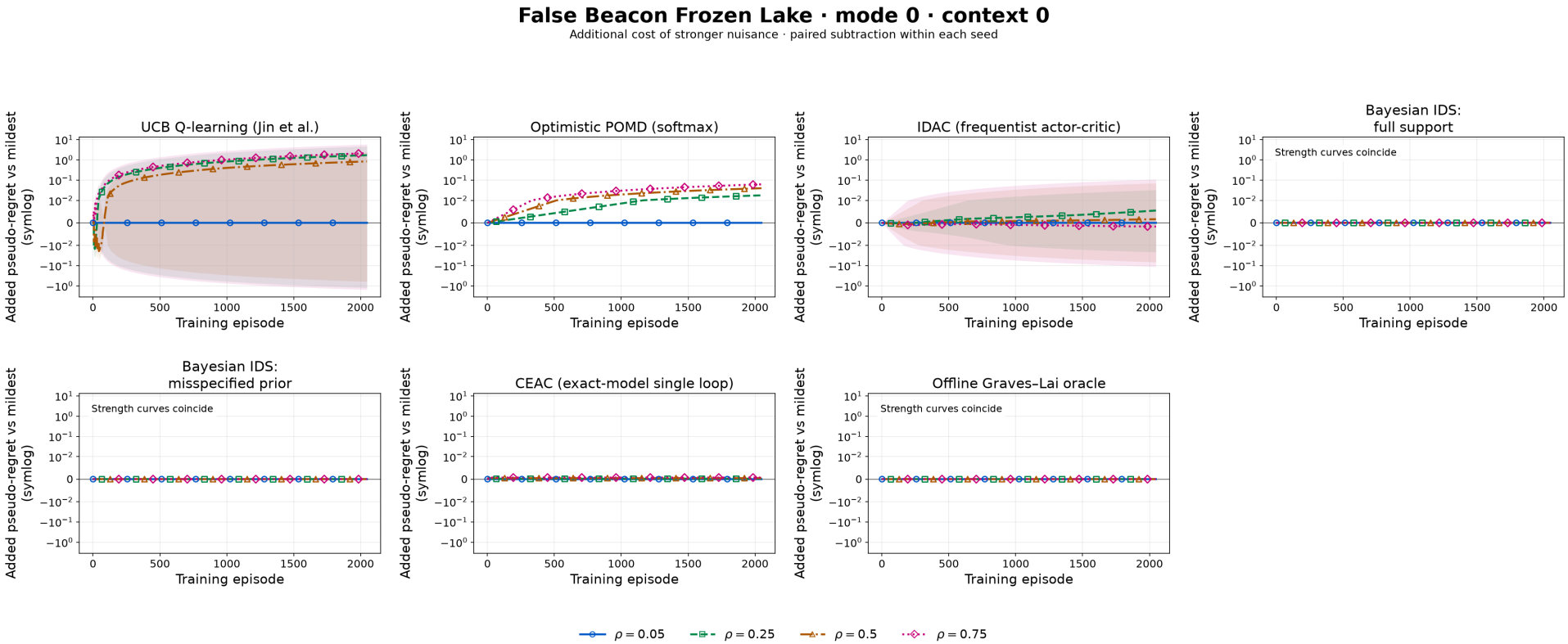


Figure 19: pathology added regret frozenlake context0 mode0

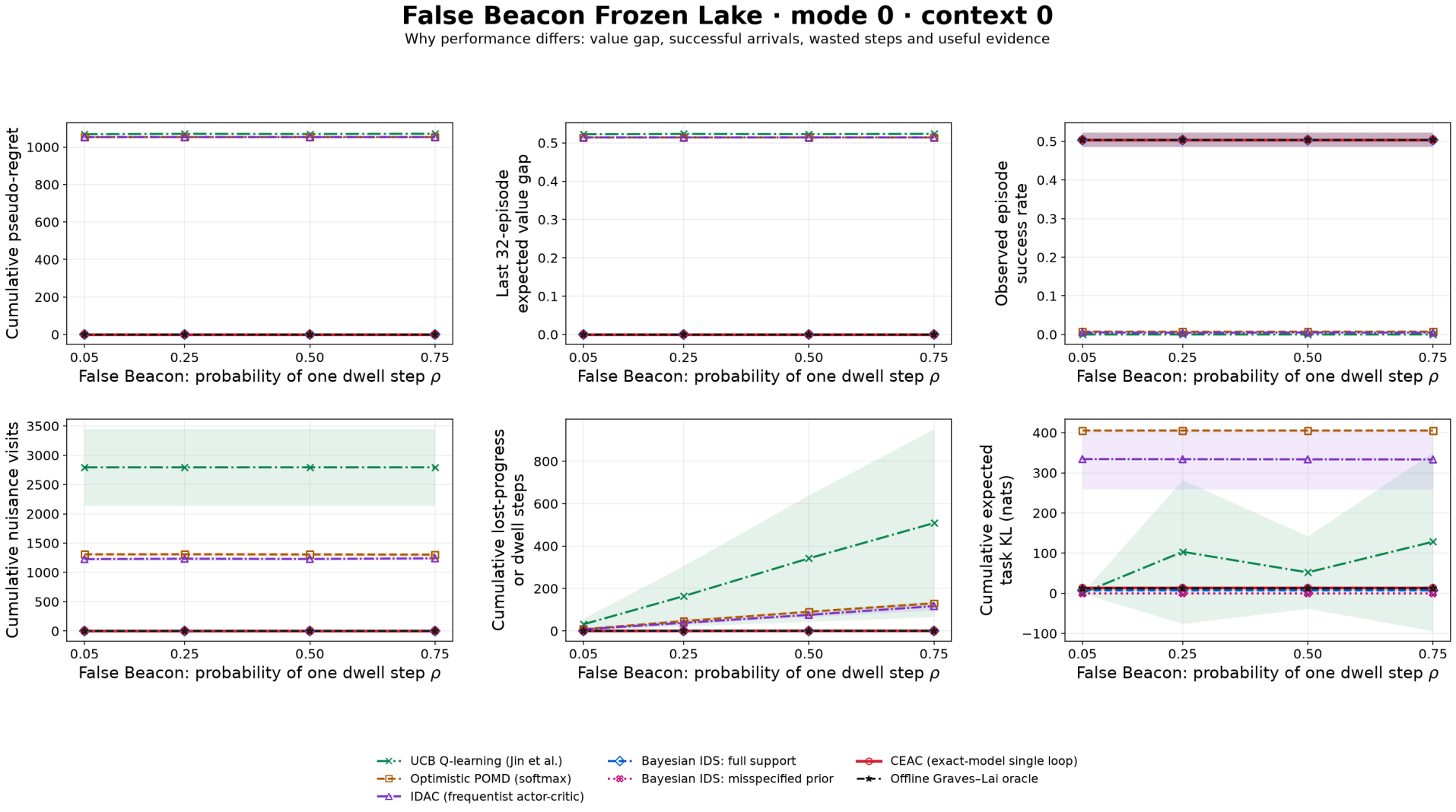


Figure 20: pathology diagnostics frozenlake context0 mode0

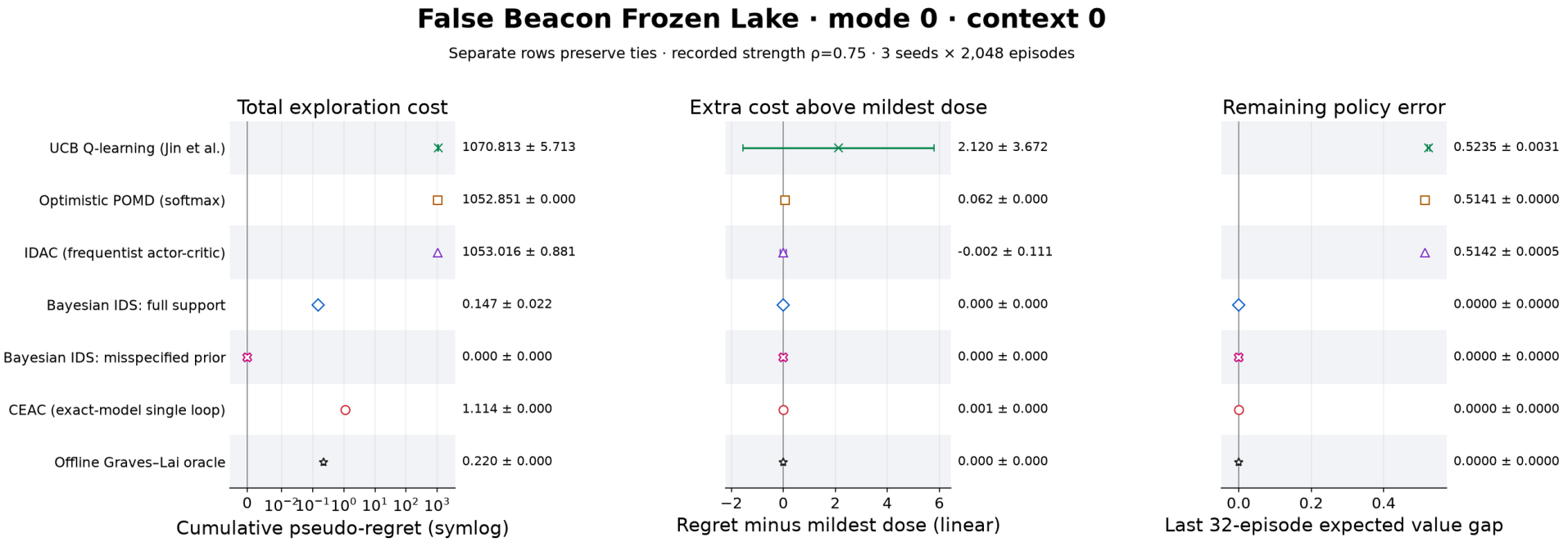


Figure 21: pathology endpoints frozenlake context0 mode0

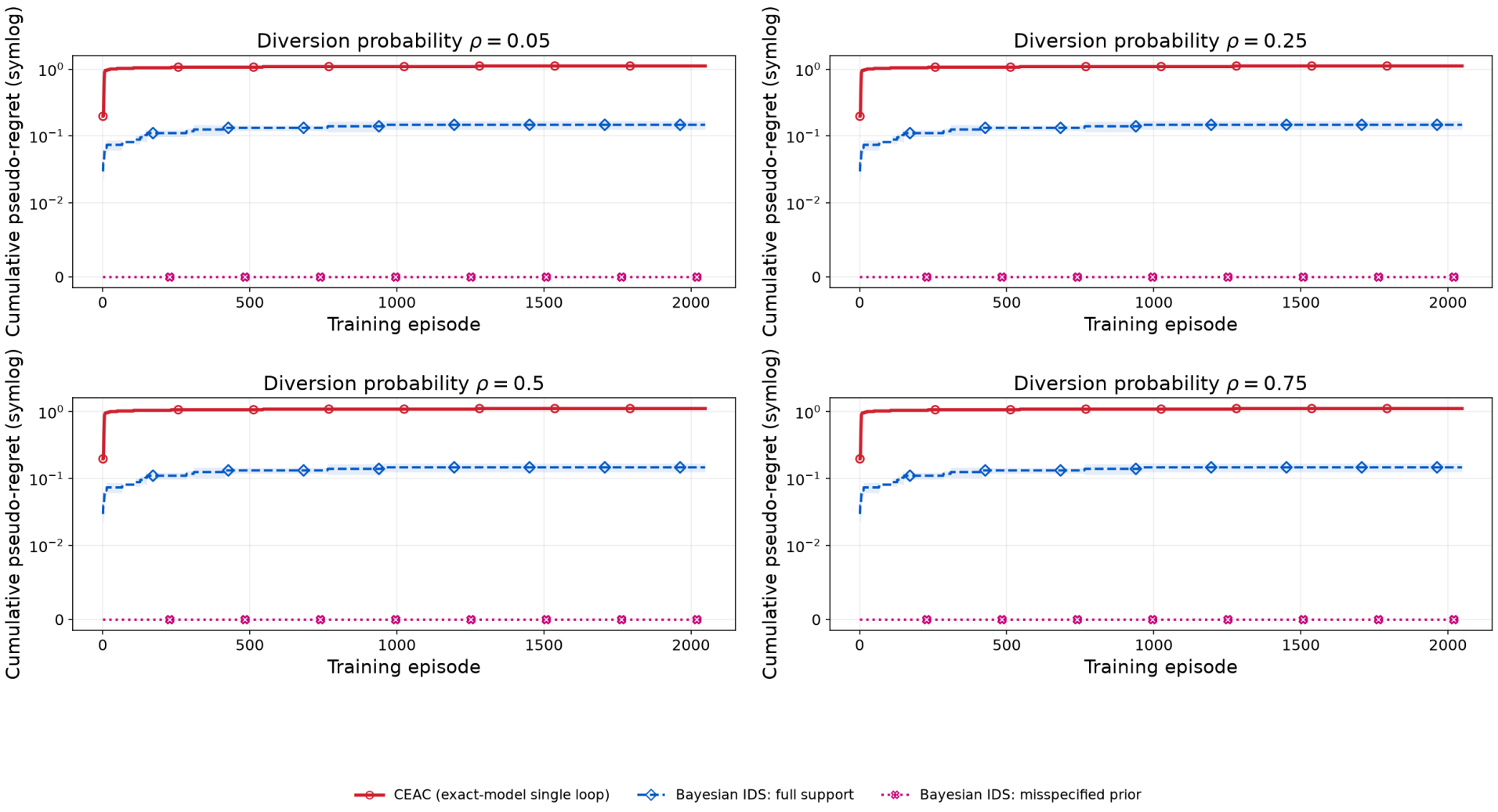


Figure 22: pathology prior frozenlake context0 mode0

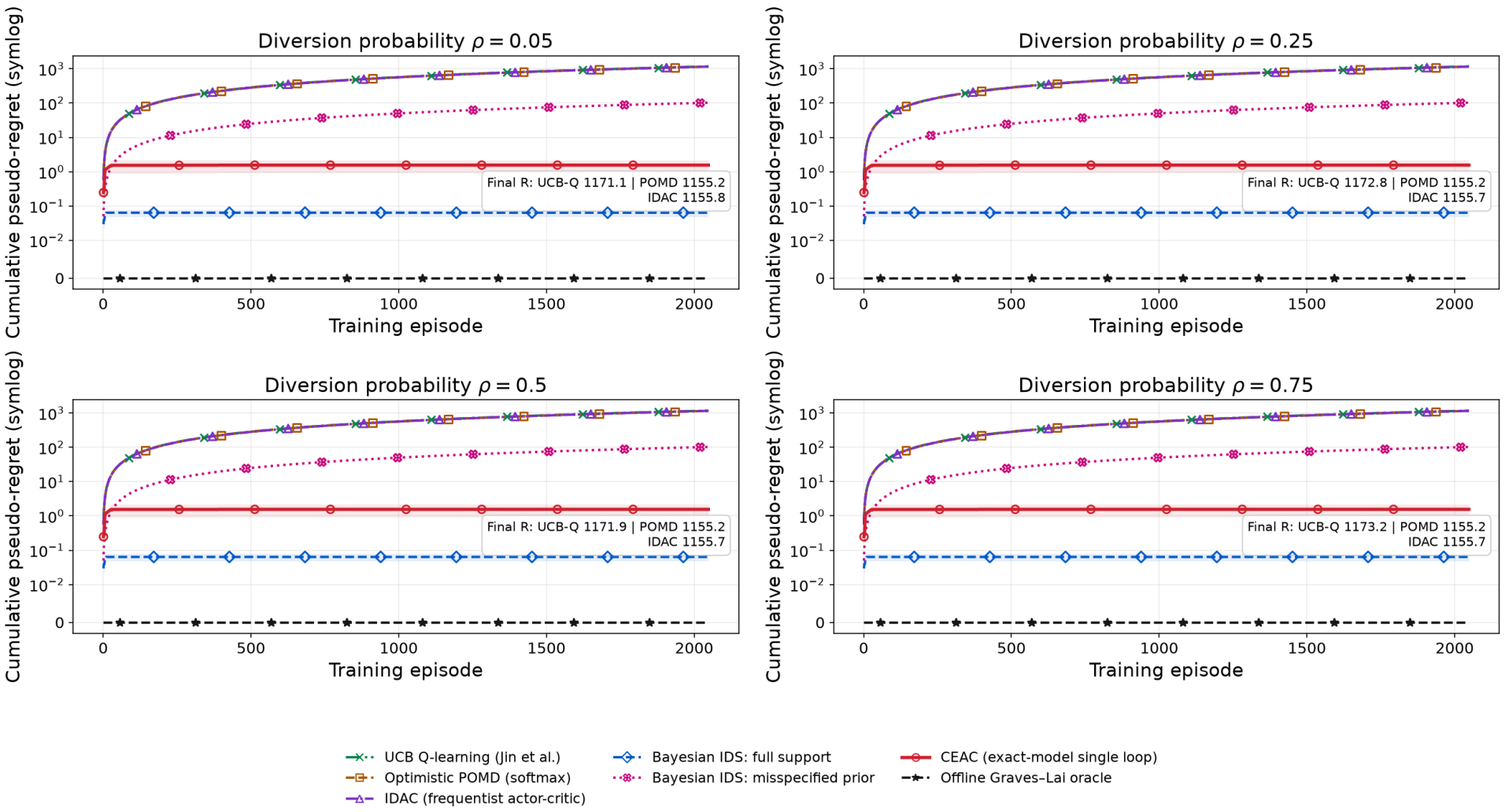


Figure 23: pathology overlay frozenlake context0 mode1

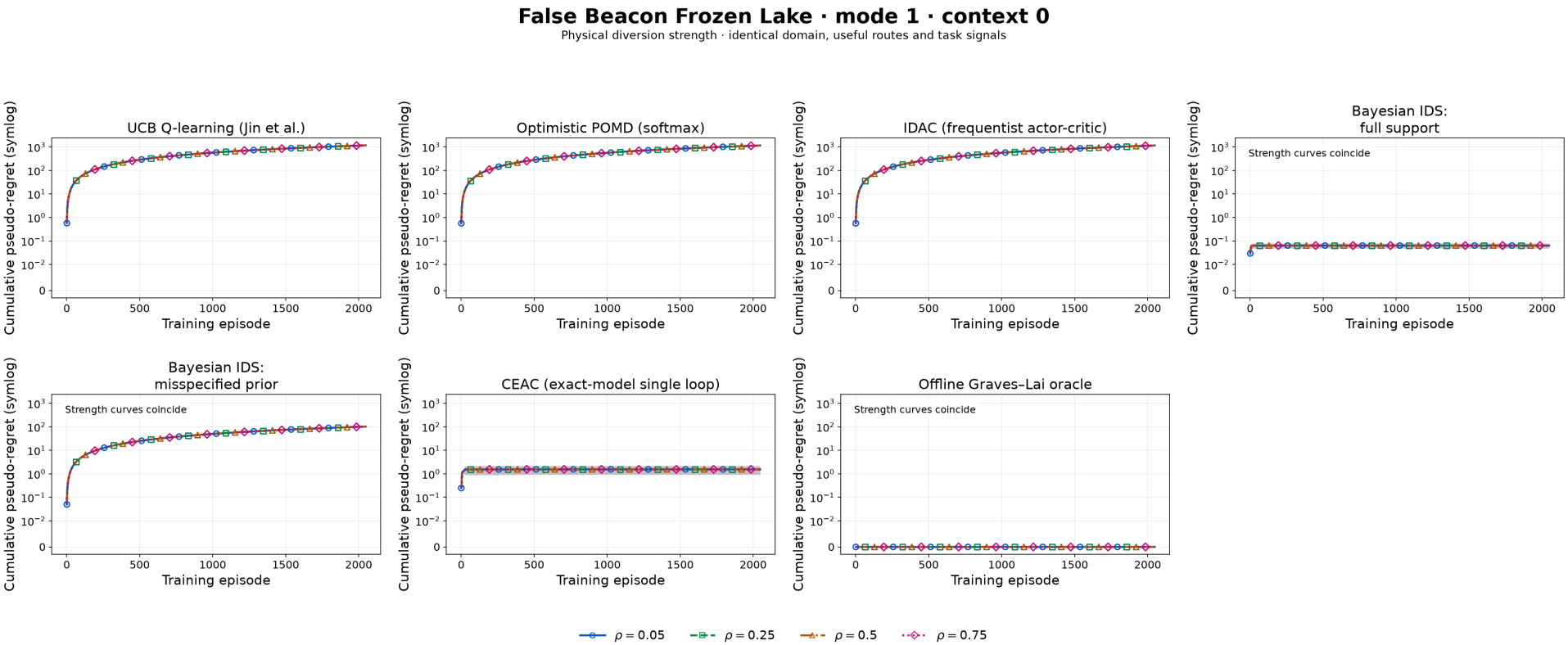


Figure 24: pathology learning frozenlake context0 mode1

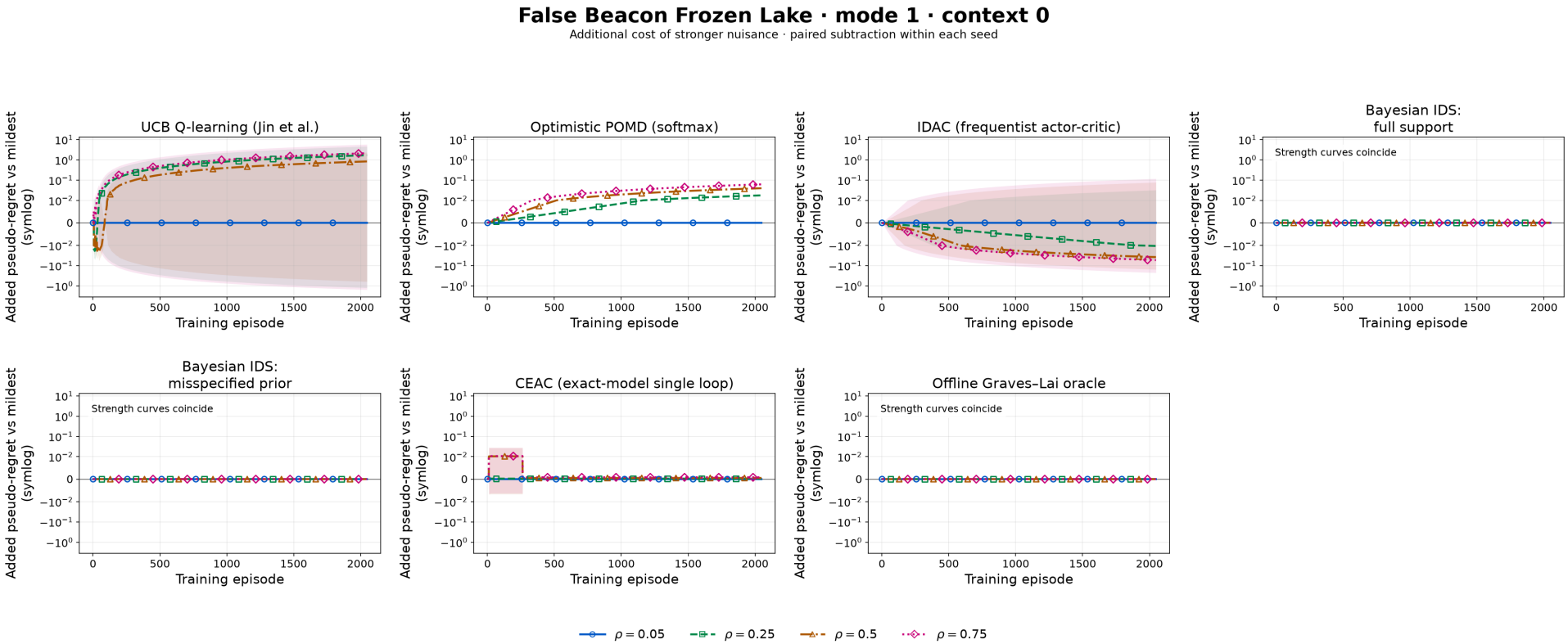


Figure 25: pathology added regret frozenlake context0 mode1

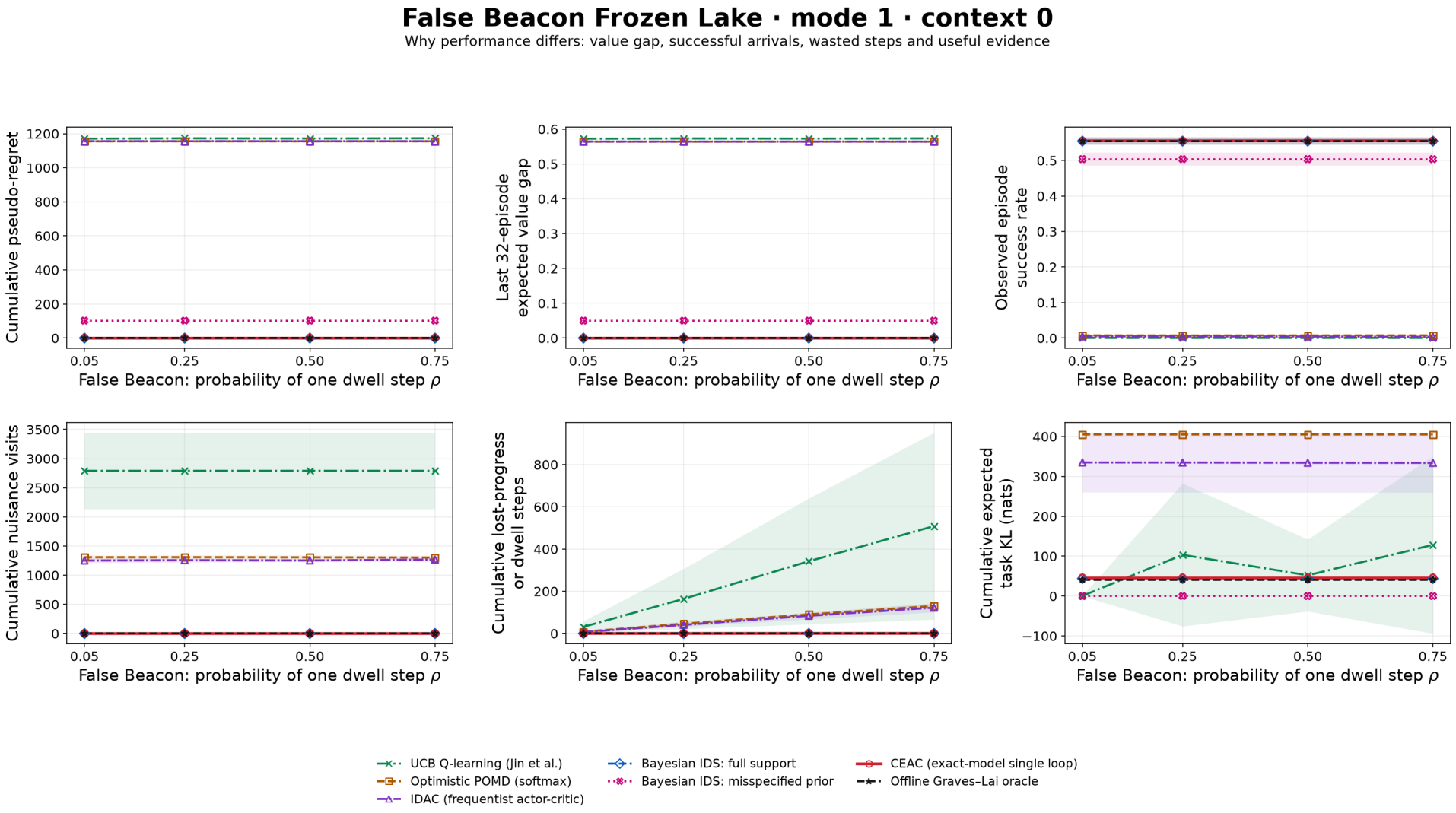


Figure 26: pathology diagnostics frozenlake context0 mode1

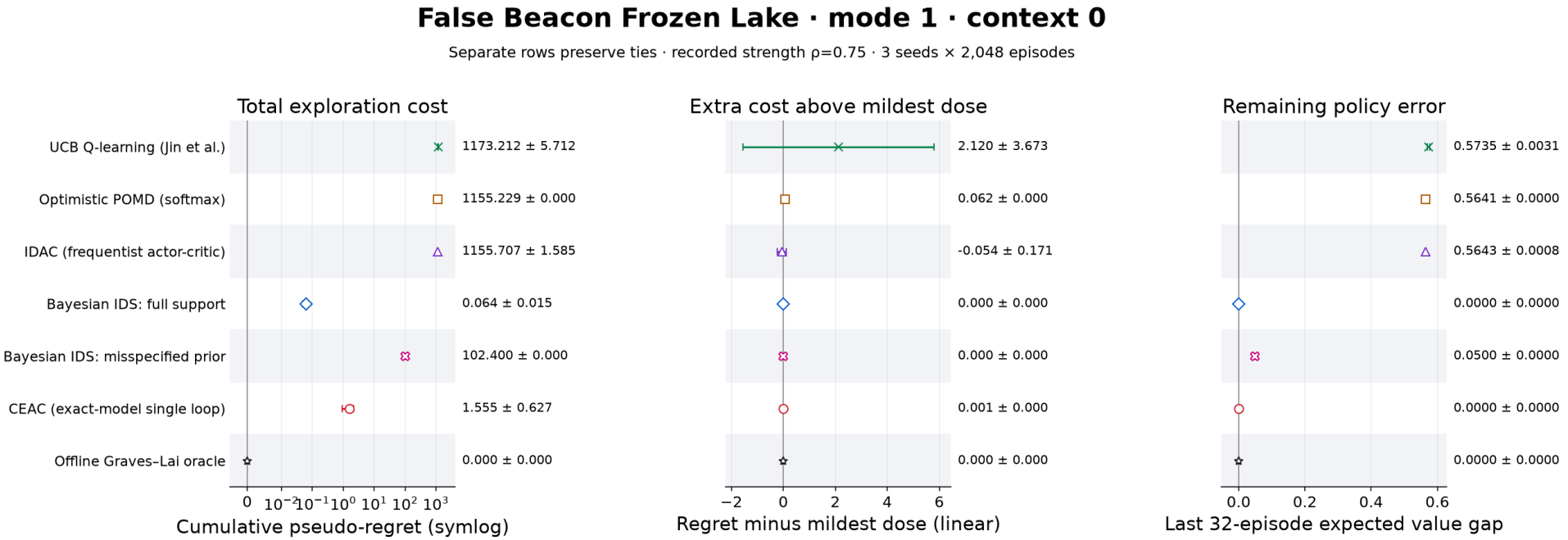


Figure 27: pathology endpoints frozenlake context0 mode1

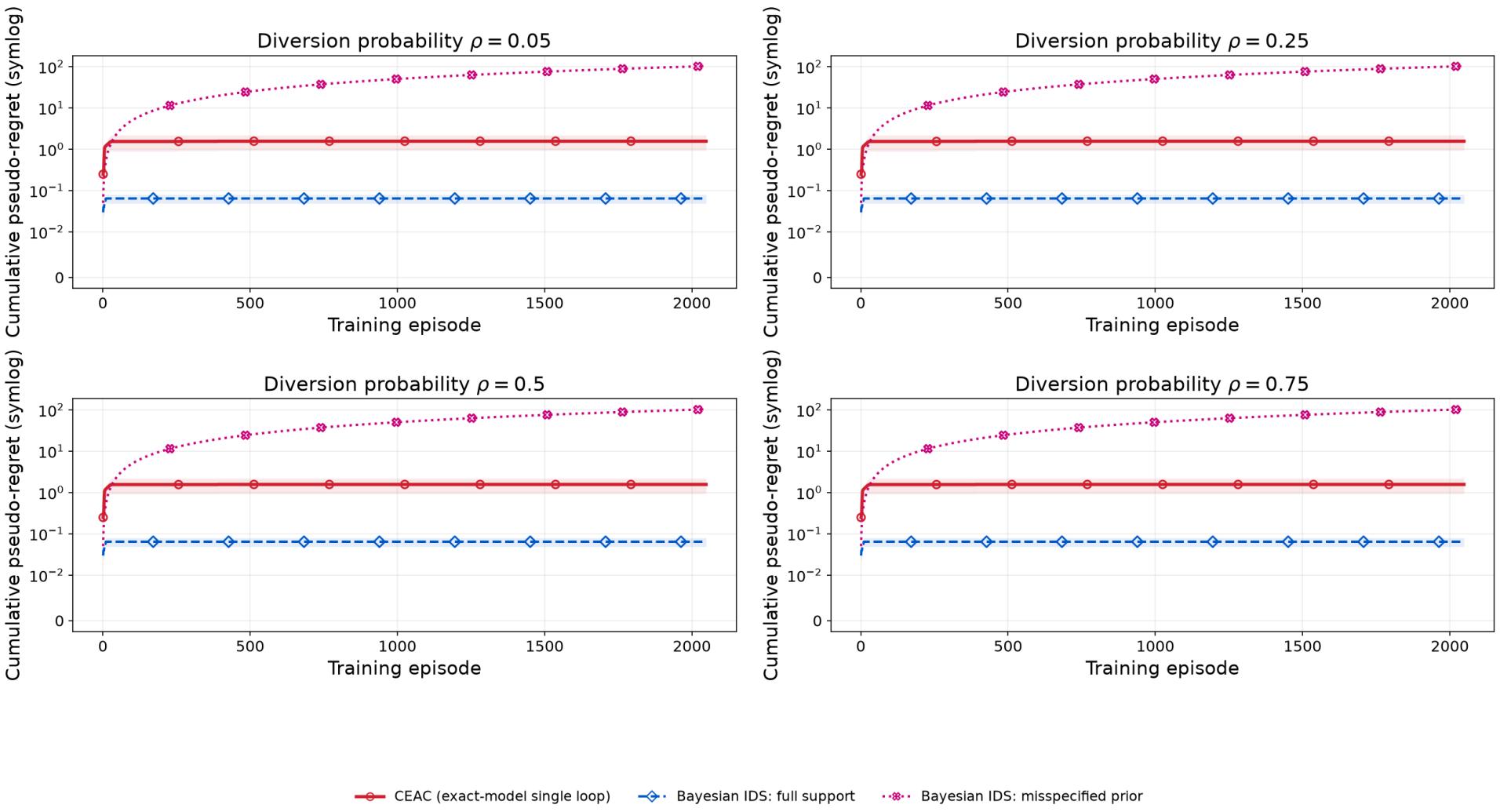


Figure 28: pathology prior frozenlake context0 mode1